GIES MSBA WORKSHOP SERIES · WORKSHOP 4

ML Foundations I

Today we cover supervised learning: teach the research agent to forecast a number, measure its error, improve it, and test it on years it never saw.

Chapter 4 of the interactive textbook · dl-msba-workshop.azurewebsites.net/intro.html

BEFORE WE START

Scan to open today's chapter

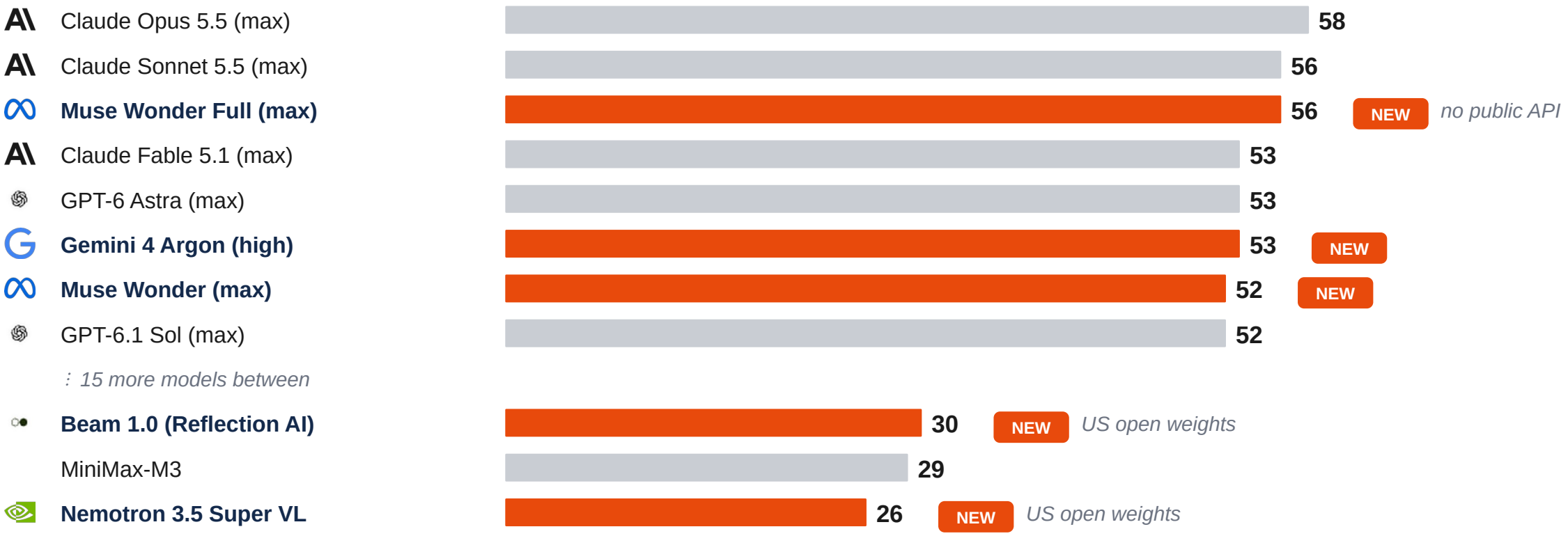


Chapter 4 · ML Foundations I

1. Scan the code with your phone camera, or open the textbook on your laptop.
2. Sign in with your @illinois.edu account (campus copy).
3. Keep it open: the notebook exercises at the end run there.

dl-msba-workshop.azurewebsites.net/intro.html

Google, Meta, Reflection and NVIDIA all shipped new models



Source: Artificial Analysis Intelligence Index v4.3.2 (10 evaluations), artificialanalysis.ai, viewed Oct 16, 2026; higher is better; 27 models shown on the site, 11 here.

What each lab shipped, and how you can use it

GOOGLE

Gemini 4 Argon

Broad access Oct 13 — **Index 53**

Flagship reasoning model for coding and science. In the Gemini API now, with a 1M-token context.

META

Muse Wonder · Wonder Full

Released Oct 12 — **Index 52 · 56**

Meta’s first frontier models. Wonder on Meta’s API at $4 / $20 per million tokens; Full only inside Muse Code.

NVIDIA

Nemotron 3.5 Super VL

Released Oct 15 — **Index 26**

Open weights; reads text, images and video. Download it and run it on your own GPUs.

For our agent: the model is a swappable part. Today’s lesson, testing on held-out data, is how you pick one.

Reflection AI’s Beam is America’s biggest open-weight bet yet

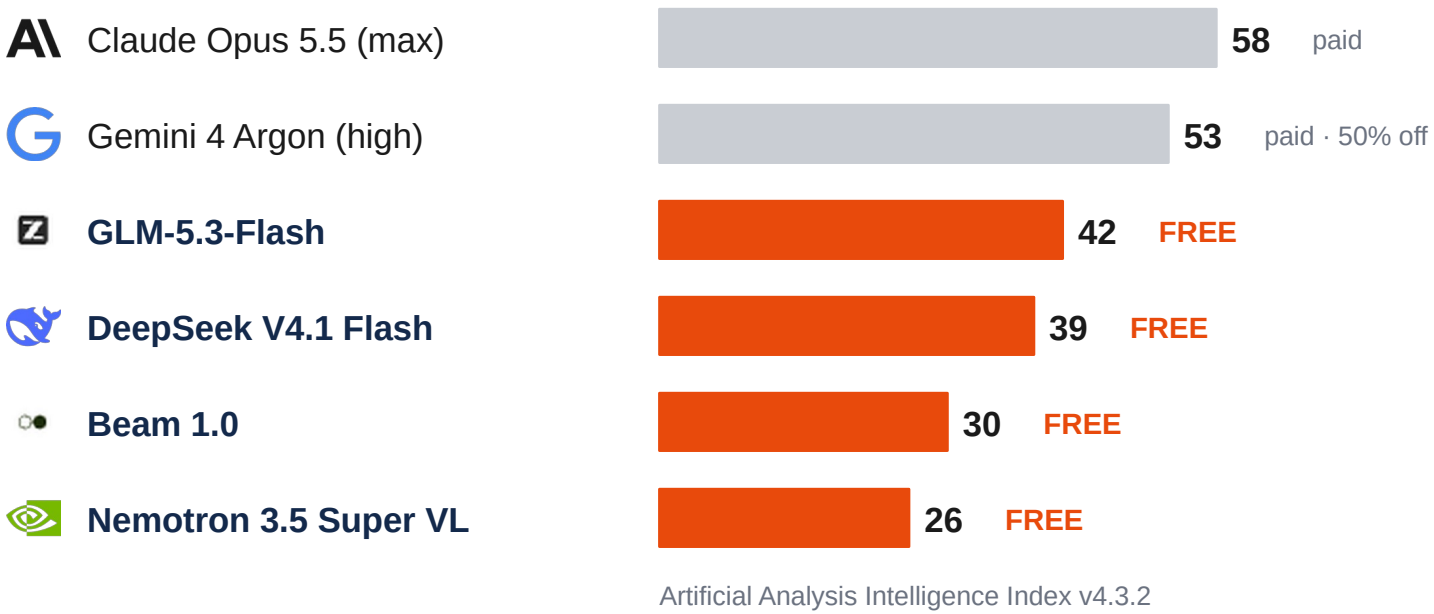
REFLECTION AI · BEAM 1.0

501B	23B	Apache 2.0	30
total parameters	active per token	licence for the weights	Intelligence Index

- Pitch: match China’s open models on reasoning with 3–4× less compute (Reflection’s own estimate)
- Same lane as NVIDIA’s Nemotron: US labs releasing weights anyone can download
- Free to try on OpenCode right now

Sources: Reflection AI, “Introducing Beam”, Oct 5, 2026; TechCrunch, Oct 5, 2026; Artificial Analysis Intelligence Index v4.3.2 (Oct 16, 2026).

OpenCode is offering strong open models free, for a limited time



THE FINE PRINT

- 15 free models; no end dates given
- Some free tiers use your prompts to improve the model
- Top closed models are still paid
- Never paste client data into a free tier

Sources: OpenCode console model catalogue (opencode.ai), viewed Oct 16, 2026; Artificial Analysis Intelligence Index v4.3.2 (Oct 16, 2026).

NVIDIA and Microsoft: laptops that run big models locally

NVIDIA RTX SPARK · WINDOWS PCS · AVAILABLE OCT 16

128 GB
unified memory (max)

1 PFLOP
of AI compute (FP4)

$2,599
starting price

- Makers: ASUS, Dell, HP, Lenovo, MSI, and Microsoft's own Surface Laptop Ultra
- Runs open models on the laptop, e.g. NVIDIA's Nemotron 3.5 Bolt (74B)
- **Why it matters: client data never has to leave the machine**

Source: NVIDIA newsroom, "NVIDIA RTX Spark PCs Available Today", Oct 16, 2026.

BEFORE WE START · MARKETS

CoreWeave is down 8% today on worries about its debt

−7.96%

CoreWeave (CRWV) $64.14, down $5.55 from Thursday's $69.69 close

Near its 52-week low of $59.51; down more than 15% this week.

WHY IT'S FALLING

1. Its 2031 bonds now yield close to 16%
2. Credit-default swaps briefly above 1,100 bp
3. Chip stocks: SOX index −12% in six sessions
4. September CPI 3.7% year on year: rate-rise fears

Today's link: "Will this borrower default?" is a classification problem. Part 7 shows the methods.

Price: CRWV quote, Oct 16, 2026. Context: Yahoo Finance and Investing.com, Oct 16, 2026. Bond, CDS and SOX figures as reported by Investing.com.

Last session: the agent learned to look things up

1 Retrieve

Find the clauses that match the question

→

2 Augment

Put them on the desk (the prompt)

→

3 Generate

Answer from them, citing the clause

FOUR WAYS TO RETRIEVE

Keyword	exact words: a ticker, a clause number
Semantic	meaning: “quiet period” finds “blackout”
Hybrid	both lists merged: where most systems start
Rerank	re-score the shortlist, best first

WORKED EXAMPLE

Tom wants to sell Deere 6 days after the firm published research on it.

Retrieved: 14-day blackout clause

Hold until day 14

Without RAG the model guesses from public text and says “yes”.

Source: course textbook, chapter 3.1 Retrieval and RAG (Champaign Capital handbook, 73 clauses). Recording: youtu.be/Rx-BJtDumqU

…to remember what a client told us

Short-term memory

This conversation. When it outgrows the desk:

Trim

Turn 1: “I prefer tables”

Turn 2

Turn 3

Turn 4

Turn 5

Early turns gone

Summarize

Summary of turns 1–3

Turn 4

Turn 5

Kept only if the summary kept it

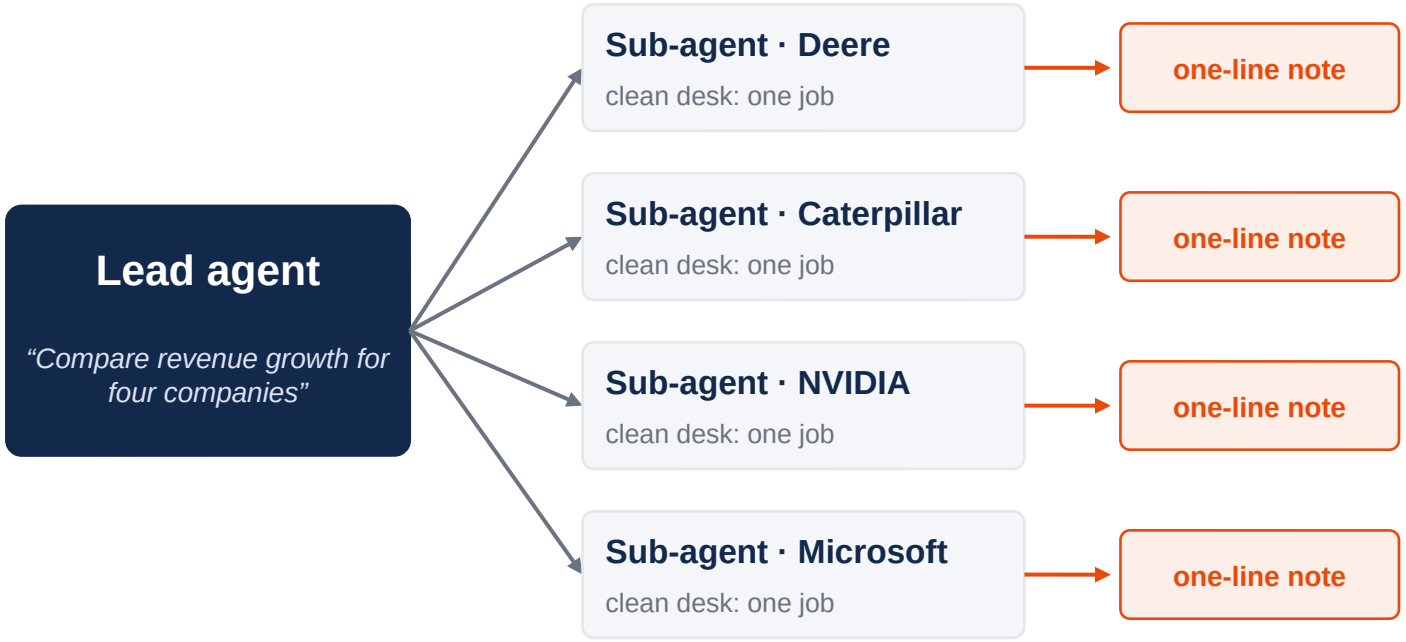
Long-term memory

Notes saved after a session, put back next time

Episodic	Semantic	Procedural
What happened	**What is true**	**How we work**
“Monday’s memo: Deere grew 6.4%”	*“Dana prefers tables”*	*“Look up both, cite each source”*
Old figures can mislead	Replace it when it changes	A person edits it

A wrong note is repeated with confidence: one saved Deere’s 6.4% as 64.0%. Gate what gets saved: save · ask a person · skip.

…and to hand independent jobs to sub-agents



SAVES

The lead reads 4 short notes, not 4 filings of ~1,800 tokens each

COSTS

9 model calls instead of 5. A sub-agent sees nothing the lead doesn’t hand it.

Use when jobs are independent and each reads a lot.

Still missing: a number the agent can defend. Today: machine learning.

Today we cover supervised learning, from data to a tested model

01	**Introduction to ML**	0–20 min
02	**Data representation**	20–40 min

Supervised

learns from labeled answers

- What it is · types
- 03 Linear regression 40–65
- 04 Loss functions 65–90
- 05 Gradient descent 90–110
- 06 Model evaluation 110–120
- 07 Ridge/lasso, classifiers, ensembles

e.g. Earnings → S&P 500 level

Unsupervised

next week · Oct 23

- What it is · types
- k-Means & Gaussian mixtures
- Hierarchical & DBSCAN
- PCA
- Anomalies & recommenders

e.g. 400 companies → groups

Reinforcement

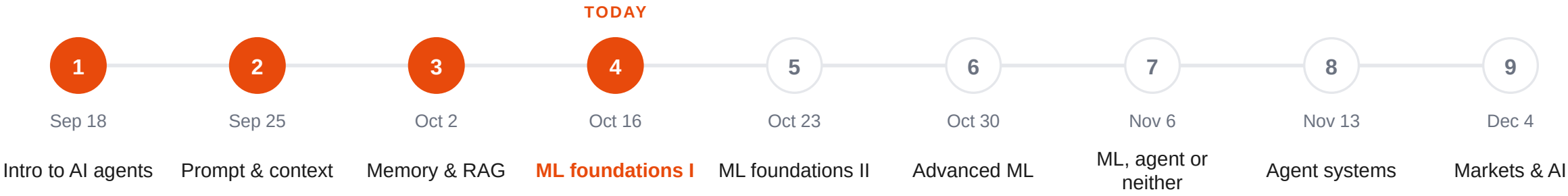
next week · Oct 23

- What it is · types
- Q-learning & DQN
- Policy gradients & PPO
- Actor-critic
- Bandits & RLHF

e.g. Sell in slices → better price

WEEK BY WEEK

Each week the agent does more; today it learns to forecast



SCHEDULE CHANGE

All of ML in two sessions was too much, so it now spreads over three:

Today · Oct 16

ML Foundations I · Supervised learning

Oct 23

ML Foundations II · Unsupervised & reinforcement learning

Oct 30

Advanced ML · Deep learning & transformers

01

Introduction to ML

0–20 min · what ML is, and which kind fits a beat-or-miss call, client groups or order execution.

The PM asks: earnings rose — where should the S&P 500 be?



THE JOB

- 44 years of index levels and earnings per share
- Give back one number, plus how wrong it might be
- A rule of thumb guesses; a model learns from the data

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Teaching example, not an investment recommendation.

Machine learning finds the rule from examples instead of being told it



“Flag any stock with P/E above 30” is a rule someone wrote. ML learns the rule, here 22.9 × earnings − 60, from 44 years of answers.

Use ML when the rule is too complex, changes over time, or needs a measured error.

Everyday: predicting an exam score from hours studied

THE STORY

Programming: a teacher writes the rule by hand. Machine learning: show the computer 100 past students and let it find the rule.

Learned rule: score ≈ 5 × hours + 50

Made-up numbers, for illustration

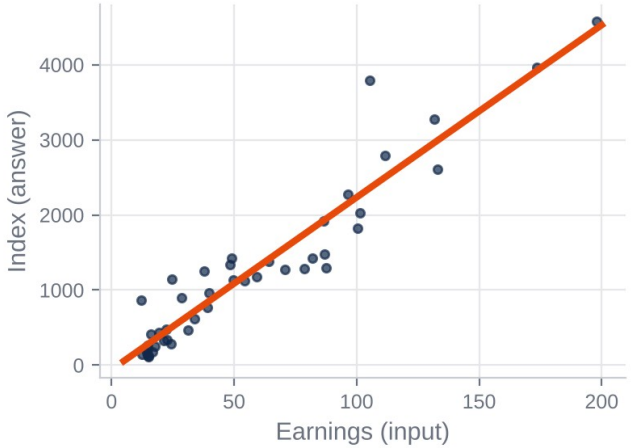
Variable	What it is	Example
Hours studied	Input: what we know	6 hours
Exam score	Answer we want to predict	80 out of 100
5 and 50	The rule, learned from data	+5 points per hour

IN OUR FINANCE EXAMPLE **Hours → earnings per share. Score → S&P 500 level. Rule → 22.9 × earnings − 60.**

What a model learns from decides which kind of ML it is

Supervised

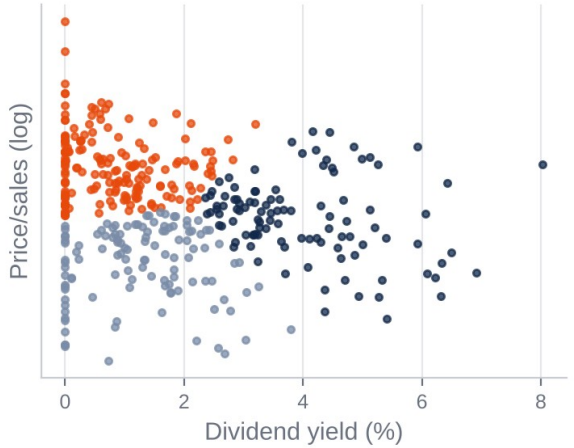
Learns from labeled answers



Earnings → index level (today)

Unsupervised

Finds structure, no labels



400 companies → 3 groups

Reinforcement

Learns from rewards for actions



Sell in slices → better price

Supervised and unsupervised panels: real data from this deck (Shiller S&P 500; S&P 500 fundamentals, Oct 16, 2026). Reinforcement: diagram.

CHECKPOINT · TAKE 30 SECONDS

A client asks: “Will Deere beat analyst estimates next quarter?” Which kind of ML fits?

CONTEXT We have 40 past Deere quarters, each with the right answer recorded: beat or miss. Supervised = learns from labeled answers. Unsupervised = no labels, finds groups. Reinforcement = learns from rewards for its own actions.

A Unsupervised: let the model find groups of quarters on its own

B Supervised: learn from past quarters labeled beat or miss

C Reinforcement: reward the model each time it trades the stock

CHECKPOINT · ANSWER

A client asks: “Will Deere beat analyst estimates next quarter?” Which kind of ML fits?

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✓ **B. Supervised: learn from past quarters labeled beat or miss**

C Reinforcement: reward the model each time it trades the stock

WHY

The past answers are known, so the model learns from labeled examples. A yes/no answer makes it classification, covered in part 7.

02

Data representation

20–40 min · turning 44 Januaries of S&P 500 earnings, dividends and Treasury yields into X and y.

S&P 500 history: one row per January, 1980–2023

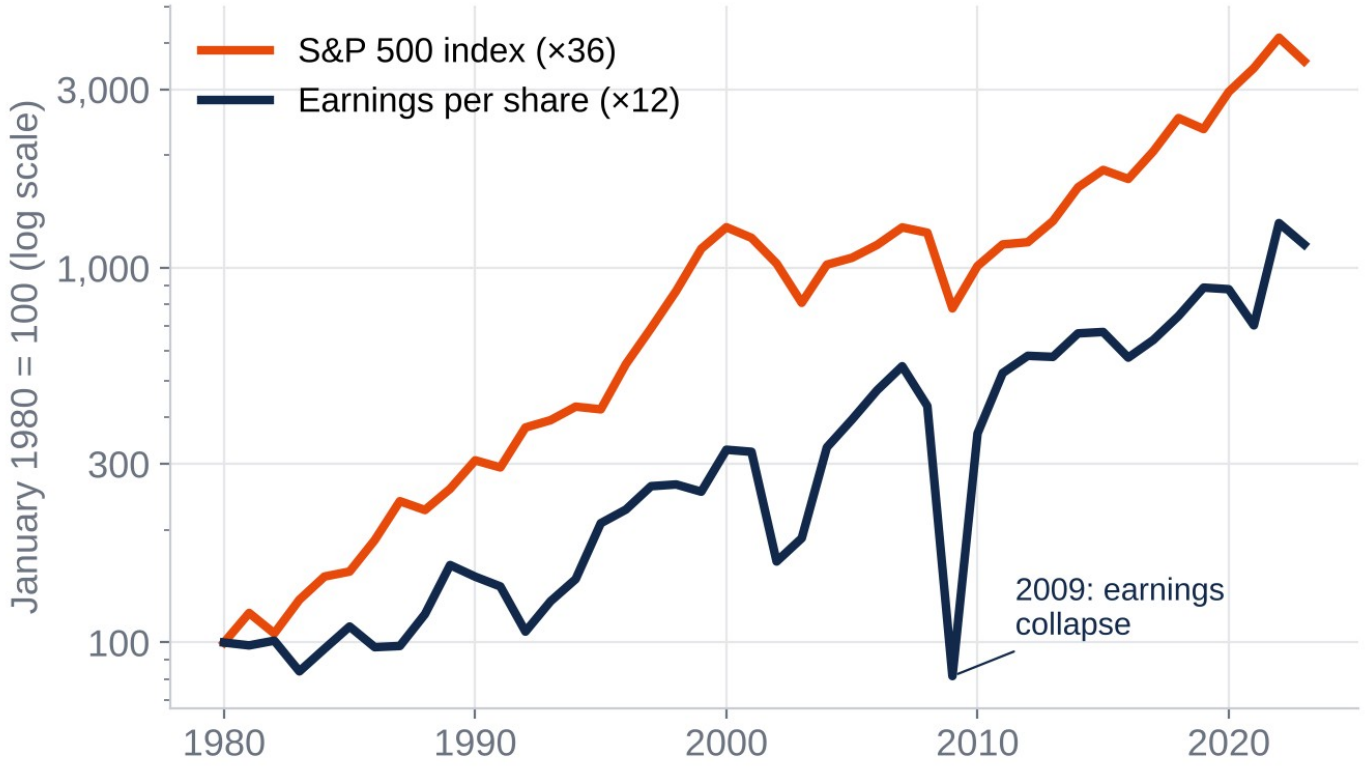
Year	S&P 500 level	Earnings / share ($)	Dividends / share ($)	10-yr yield (%)	CPI	CAPE
1980	111	15.00	5.70	10.80	77.8	8.9
1990	340	22.49	11.14	8.21	127.4	17.1
2000	1,426	49.10	16.71	6.66	168.8	43.8
2008	1,379	64.25	27.92	3.74	211.1	24.0
2009	866	12.21	28.01	2.52	211.1	15.2
2020	3,278	131.76	58.69	1.76	258.0	31.0
2022	4,574	197.88	60.92	1.76	281.2	36.9
2023	3,961	173.56	67.35	3.53	299.2	28.3

8 of 44 rows shown. Orange column = the label y; the others are candidate features. 2009 (shaded): earnings collapsed to $12.

Used in parts 2–7 · monthly rows from 1950 for the classification questions · file: sp500_shiller_january_1980_2023.csv

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Level = January average of daily closes; earnings and dividends = trailing 12 months.

Before building any model, get to know the data



FOUR QUESTIONS FIRST

- Where is it from? Shiller's S&P 500 file, monthly since 1871
- What is one row? One January: average price, trailing 12-month earnings
- When is each value known? Never use the future
- What looks wrong? Earnings = 0 from July 2023 on

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Index and earnings rebased to January 1980 = 100.

Everyday: check the gradebook before predicting anything

THE STORY

If absent students are recorded as a 0, the model learns that studying barely matters. Look at the data first.

A missing grade is not a 0

Made-up numbers, for illustration

Check	Question to ask	Gradebook
Rows	What is one row?	One student, one exam
Columns	What does each mean? Units?	Hours per week; score out of 100
Gaps	Is missing shown as 0?	Absent = 0, should be blank
Range	Anything impossible?	A score of 130

IN OUR FINANCE EXAMPLE **S&P earnings show 0 since mid-2023: missing, not zero.**

CHECKPOINT · TAKE 30 SECONDS

The file shows S&P 500 earnings of 0 for every month since mid-2023. What do you do?

CONTEXT Shiller's S&P 500 file has 0.0 in the earnings column for every month from July 2023 to September 2026 (39 months). The index itself kept rising over that time.

A Use it: company earnings fell to zero

B Treat it as missing: not yet filled in, so leave those months out

C Fill in last year's value and move on

CHECKPOINT · ANSWER

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A Use it: company earnings fell to zero

✓ **B. Treat it as missing: not yet filled in, so leave those months out**

C Fill in last year's value and move on

WHY

Zero here means 'not entered yet'. Feeding it in as real would wreck the model, so check how gaps are coded before you build.

ONE ROW = ONE YEAR

Features are what we know; the label is what we want to predict

Year (January)	Feature x: earnings per share	Label y: S&P 500 level
1990	$22.49	340
2000	$49.10	1,426
2010	$54.29	1,124
2020	$131.76	3,278
2023	$173.56	3,961

5 of 44 rows. Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44.

Everyday: one row per student, one column per fact

THE STORY

Features are what you know before the exam. The label is the score you want to predict.

One row = one student

Made-up numbers, for illustration

Column	Role	Example row
Hours studied	Feature	6
Hours slept	Feature	7
Classes attended	Feature	12 of 14
Exam score	Label	80

IN OUR FINANCE EXAMPLE **Features: earnings, dividends, 10-year yield. Label: the S&P 500 level.**

A dataset is a matrix X of features and a vector y of labels



Add dividends and the 10-year yield as columns: X becomes 44 × 3.

Everyday: a class of 30 students becomes X and y

THE STORY

Stack every student's features into one table: that is X. Put their scores in one column: that is y.

X: 30 rows × 3 columns · y: 30 scores

Made-up numbers, for illustration

Name	What it is	Class example
X	Matrix: rows = students, columns = features	30 × 3
y	Vector: one score per student	30 numbers
One row	One student	6 h, 7 h, 12 classes → 80

IN OUR FINANCE EXAMPLE **Our X: 44 Januaries × earnings. Our y: 44 index levels.**

Alternative data: images, text and audio all become columns in X

Source	Example for an analyst	What reads it	Features it adds
Satellite & geospatial	Cars in retail parking lots; crop health across the farm belt	Image model (CNN, vision transformer)	Car counts, vegetation index
Text	Earnings-call transcripts, 10-K filings, news	LLM / language model	Tone score, risk words, guidance changes
Audio & video	Executives speaking on earnings calls	Speech-to-text + audio model	Transcript, hesitation, vocal stress
Transactions & web	Card spending, app downloads, web traffic, ship tracking	Aggregation over time	Weekly sales and demand proxies

Once encoded as numbers, it’s the same regression, classification or clustering as today.

Raw signals go through an encoder before any model sees them



Multimodal models such as Nemotron 3.5 Super VL read text, images and video in one pass.

Check before buying: cost, licence and privacy, history length, and when each value was really known.

Everyday: a photo of the library becomes one number

THE STORY

Models need numbers. An encoder turns a photo, a review or a recording into a number that becomes one more column in X.

Made-up numbers, for illustration

Raw signal	Encoder does	New column
Photo of the library	Count the people	42 students
Course review text	Score how positive it is	+0.7
Lecture recording	Measure the pace	150 words/min

IN OUR FINANCE EXAMPLE **Satellite photos → cars in parking lots → one more column next to earnings.**

TERM

Parameters: $\hat{y} = w \cdot x + b$

x the feature, read from the data (earnings)

ŷ the prediction (index level)

w the weight: slope of the line, learned

b the intercept: where the line starts, learned



WHY IT MATTERS

A P/E multiple is a parameter set by judgment. ML sets w and b from the data and can say how well they fit.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44.

Everyday: w and b in the exam-score rule

THE STORY

The rule has two knobs. Nobody types them in; training picks them from past students.

$$\hat{y} = 5 \times 6 + 50 = 80$$

Made-up numbers, for illustration

Symbol	Meaning	Exam example
x	Feature: hours studied	6
w	Weight: points per extra hour	5
b	Baseline score at 0 hours	50
ŷ	Predicted score	80

IN OUR FINANCE EXAMPLE **x = earnings, w = 22.9, b = −60, ŷ = predicted index.**

CHECKPOINT · TAKE 30 SECONDS

In ŷ = w · x + b for our index model, which part does training choose?

CONTEXT x = S&P 500 earnings per share, read from the data. ŷ = the predicted index level, calculated. w = slope of the line. b = where the line starts. Training is fitting the line to past years.

A x, the earnings figure

B w and b, the weight and the intercept

C ŷ, the prediction

CHECKPOINT · ANSWER

In ŷ = w · x + b for our index model, which part does training choose?

CONTEXT x = S&P 500 earnings per share, read from the data. ŷ = the predicted index level, calculated. w = slope of the line. b = where the line starts. Training is fitting the line to past years.

A x, the earnings figure

✓ **B. w and b, the weight and the intercept**

C ŷ, the prediction

WHY

x comes from the data and ŷ is calculated. Training picks the parameters: w and b.

FAMILY 1 OF 3 · PARTS 03–07 · TODAY'S TOPIC

Supervised learning

Learn from past examples with the answer attached. First what it is and its types, then one model end to end.

1	**Supervised**	now · labeled answers
2	**Unsupervised**	no labels
3	**Reinforcement**	rewards

SUPERVISED LEARNING

Supervised: learn from past answers, then answer a new case

1 Labeled history

Inputs plus the right answer

→

2 Learn the rule

What links input to answer

→

3 Answer a new case

New input in, prediction out

WHAT IT NEEDS

Past examples with the right answer attached: the label.

AT CHAMPAIGN CAPITAL

Forecast the S&P 500 from earnings (today).
Call Deere’s next quarter: beat or miss (part 7).

Everyday: last year’s students, with their grades, teach the model

THE STORY

Every past student comes with the answer: their real score. The model learns from them, then predicts a new student it has never seen.

New student: 6 hours → predicted 80

Made-up numbers, for illustration

Variable	Role	Example
Hours studied	Input (feature)	2, 4, 6 …
Exam score	Answer (label), known for past students	60, 70, 80 …
New student’s score	What the model predicts	?

IN OUR FINANCE EXAMPLE **Past Januaries, each with the index level known → predict the index from earnings.**

Supervised: predict a number or a category

Regression

Predicts a number

- Linear regression **TODAY**
- Ridge regularization
- Lasso regularization

FINANCE USE

Forecast the S&P 500 level from earnings, dividends and the 10-year yield.

Classification

Predicts a category

- Logistic regression
- k-Nearest Neighbors
- Support Vector Machines
- Decision trees

FINANCE USE

Will Deere beat analyst estimates next quarter: yes or no?

Ensemble methods

Many models combined

- Random forests
- Gradient-boosted trees

FINANCE USE

Score credit risk on a loan or bond from dozens of borrower features.

Everyday: predict the exact score, or just pass or fail?

THE STORY

Same students, same inputs. What changes is the kind of answer you want back.

Regression → 80 points · Classification → Pass

Made-up numbers, for illustration

Type	Answer is…	Exam example
Regression	a number	Score: 80
Classification	a category	Pass or fail
Ensemble	many models voting	3 tutors: pass, pass, fail → pass

IN OUR FINANCE EXAMPLE **Index level = regression. Beat or miss estimates = classification.**

03

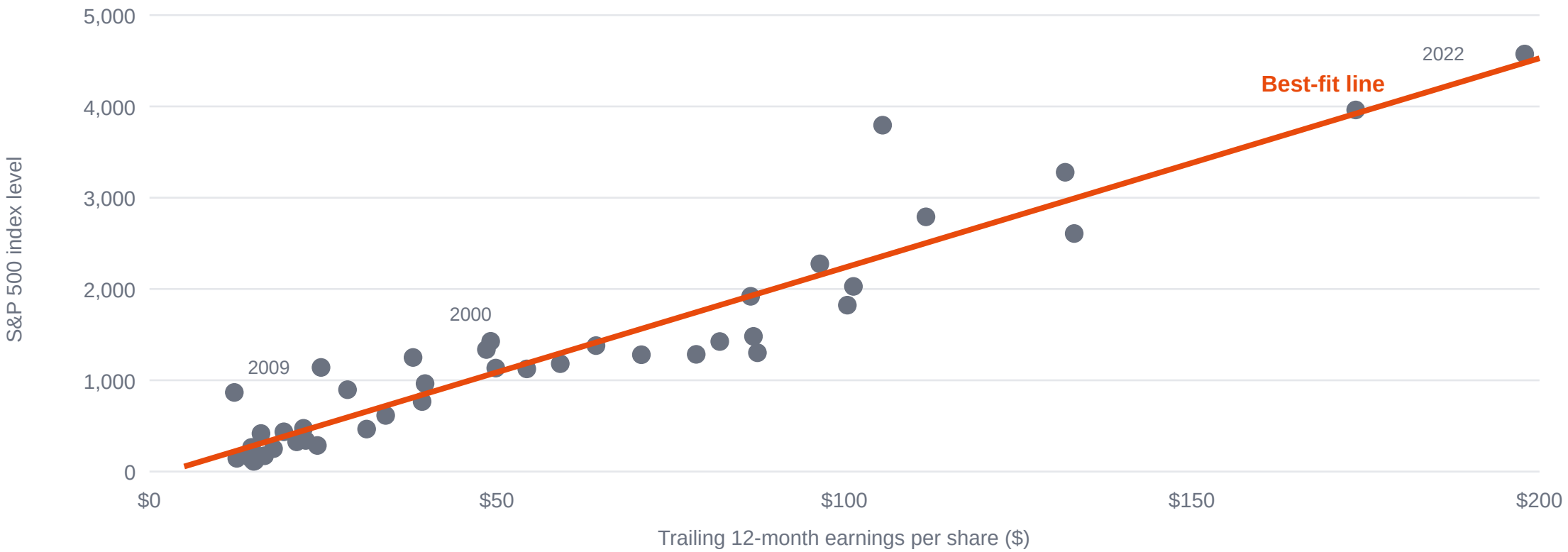
Linear regression

40–65 min · the line that turns earnings per share into an S&P 500 forecast.

SUPERVISED LEARNING

44 JANUARIES, 1980–2023

Higher earnings go with a higher index — but not perfectly



Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Line fitted by least squares.

THE FITTED LINE: $\hat{Y} = 22.9 \cdot \text{EARNINGS} - 60$

22.9

Each extra $1 of earnings per share goes with about 23 more index points.
The line explains 90% of the variation ($R^2 = 0.90$).

Least-squares fit on all 44 rows.

A prediction is just plugging a new x into the fitted line

1. Read the feature: trailing earnings per share = $173.56.
2. Multiply by the weight: 22.94 × 173.56 = 3,981.
3. Add the intercept: 3,981 − 60 = 3,921.
4. Compare with the label: actual index = 3,961.

Predicted 3,921 vs actual 3,961 — off by 40 points, about 1%.

Everyday: the best straight line through past students

THE STORY

Plot hours against score for every student. Linear regression draws the one straight line that misses them least.

Each extra hour → +5 points

Made-up numbers, for illustration

Part	What it is	Exam example
x axis	Hours studied	0 to 8
y axis	Exam score	50 to 90
Slope (w)	How steep the line is	+5 per hour
Intercept (b)	Score at 0 hours	50

IN OUR FINANCE EXAMPLE **Earnings on the x axis, index on the y axis, slope 22.9.**

SUPERVISED LEARNING

CHECKPOINT · TAKE 30 SECONDS

GIES

Earnings per share reach $200. What does ŷ = 22.9 × earnings − 60 predict?

> **CONTEXT** Our fitted line from 44 Januaries, 1980–2023: predicted index = 22.9 × earnings per share − 60. To predict, multiply by 22.9, then subtract 60.

A About 4,520

B About 4,580

C About 3,961

Earnings per share reach $200. What does ŷ = 22.9 × earnings − 60 predict?

CONTEXT Our fitted line from 44 Januaries, 1980–2023: predicted index = 22.9 × earnings per share − 60. To predict, multiply by 22.9, then subtract 60.

✓ **A. About 4,520**

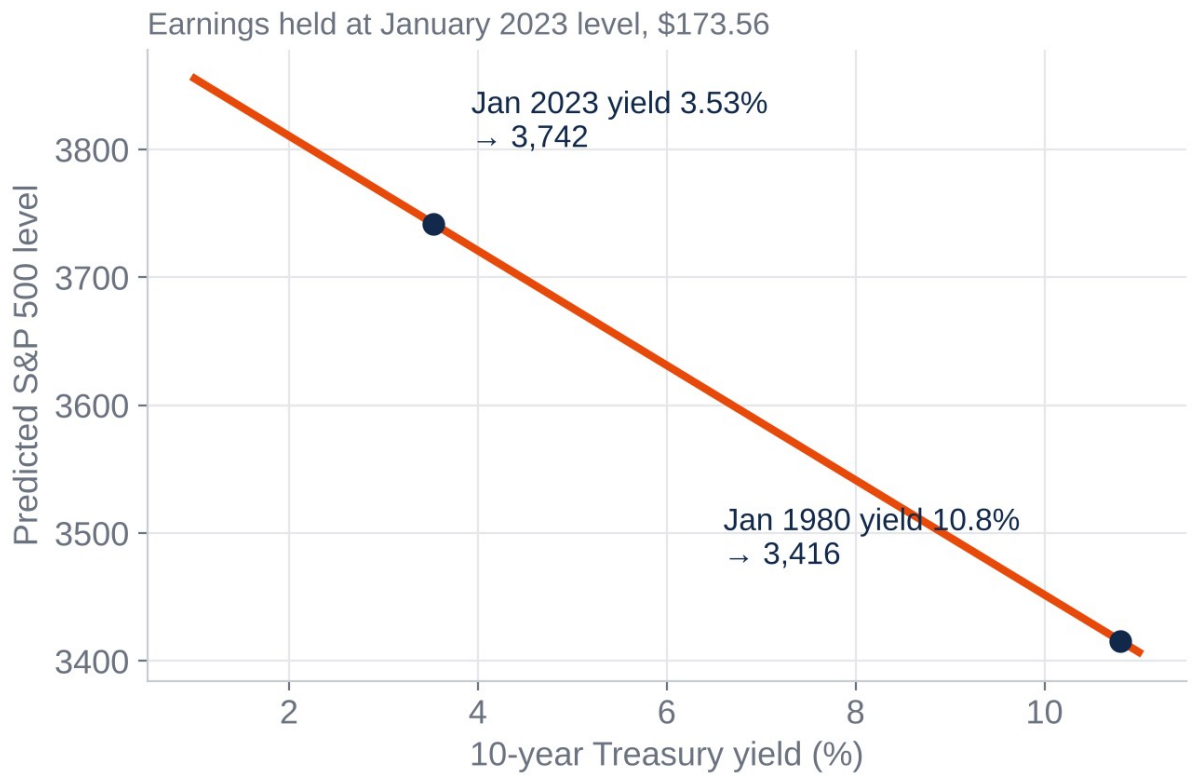
B About 4,580

C About 3,961

WHY

22.9 × 200 = 4,580, then subtract the 60 intercept: 4,520. B forgot the intercept; C is January 2023's actual level.

Higher yields, lower index: −45 points per +1 point of yield



THE FITTED MODEL

- ŷ = 20.6 · earnings − 44.9 · 10-yr yield + 327
- Each weight: one feature's effect, others held fixed
- Same loss, same training loop: ŷ = X·w + b
- Fitted on all 44 Januaries; part 6 tests it

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Least squares on earnings and the 10-year yield.

04

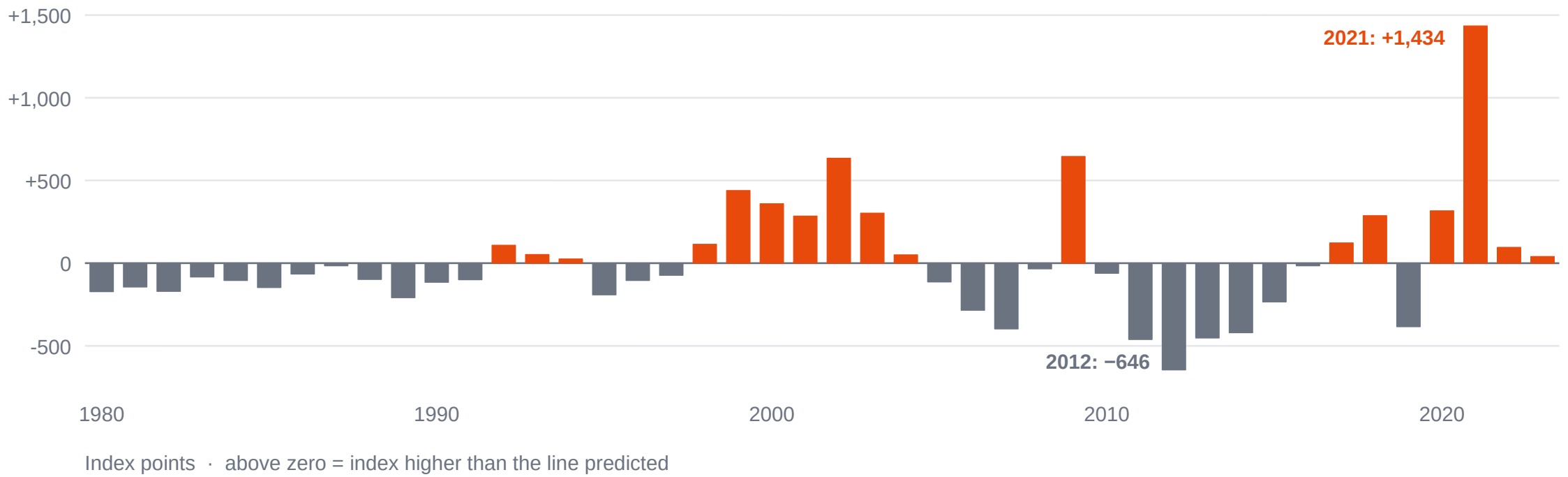
Loss functions

65–90 min · how far the forecast misses — dot-com, 2009, COVID — in one number.

SUPERVISED LEARNING

RESIDUAL = ACTUAL − PREDICTED

The line misses most in bubbles and crashes



Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44.

Square each miss, then average: that is mean squared error

1. Residuals: 2000 = +359, 2012 = −646, 2023 = +40.
2. Square them: 129,114 · 417,451 · 1,567. Signs disappear.
3. Average: (129,114 + 417,451 + 1,567) ÷ 3 = 182,711.
4. Square root to get back to index points: √182,711 ≈ 427.

Over all 44 years: MSE = 121,261, so RMSE = 348 index points.

Everyday: how wrong were our three score guesses?

THE STORY

Predicted vs actual for three students. Each miss is a residual. Square each miss, then average.

MSE = (64 + 16 + 4) ÷ 3 = 28 · RMSE ≈ 5.3 points

Made-up numbers, for illustration

Student	Predicted → actual	Miss → squared
Ana	80 → 72	8 → 64
Ben	70 → 74	−4 → 16
Cara	60 → 62	−2 → 4

IN OUR FINANCE EXAMPLE **RMSE 348: the line's typical miss is 348 index points.**

SUPERVISED LEARNING

CHECKPOINT · TAKE 30 SECONDS

GIES

Over 10 years, A misses by 100 points every year; B is perfect except one 1,000-point miss. Lower MSE?

CONTEXT MSE = the average of each miss squared. MAE = the average size of each miss. A: ten misses of 100. B: nine misses of 0 and one of 1,000. Both have MAE = 100.

A Model A

B Model B

C They tie

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✓ **A. Model A**

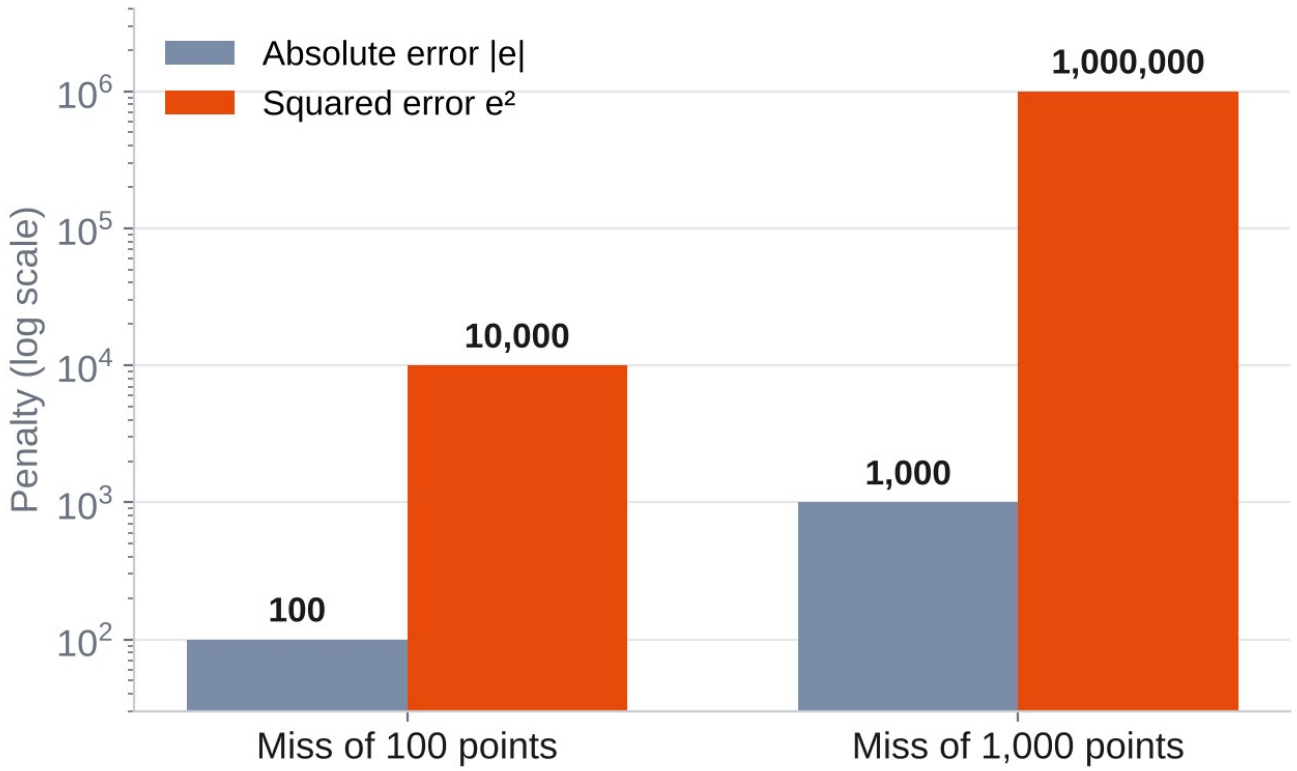
B Model B

C They tie

WHY

A: $100^2 = 10{,}000$ each year. B: $1{,}000^2 \div 10 = 100{,}000$. Squaring makes one big miss count ten times more.

Squared error punishes big misses far more than absolute error



ON OUR LINE

- MSE / RMSE: big errors dominate. RMSE 348
- MAE: robust to a few outliers. MAE 241
- Risk desk: one huge miss hurts most, so square it
- Screening many stocks: absolute error is steadier

Penalty bars are arithmetic (100², 1,000²). RMSE and MAE: Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44.

± 348

A typical miss of the line, in index points. Report it with every forecast.

RMSE over 44 Januaries, in-sample. Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44.

05

Gradient descent

90–110 min · how the model walks to the earnings multiple that fits best.

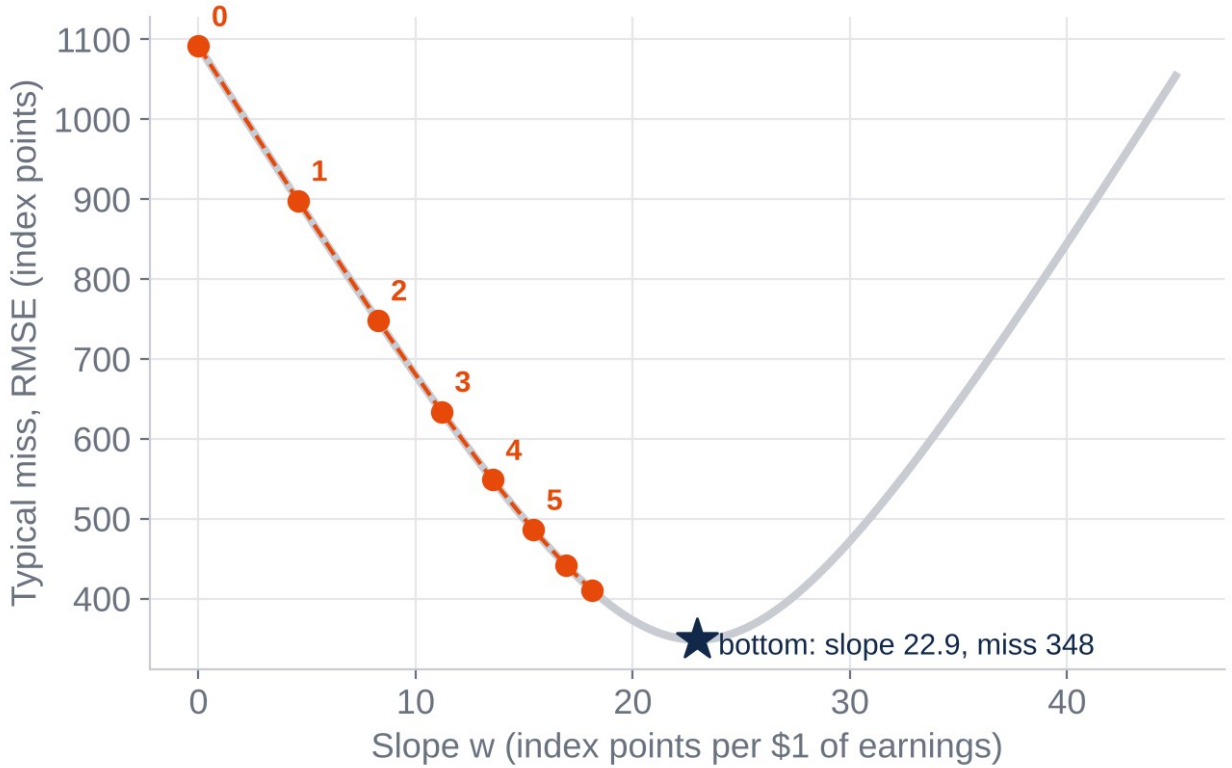
Error is lowest at a slope of about 23, and rises on both sides



Training means walking to the bottom of this valley.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Intercept re-centred for each slope.

Gradient descent: feel which way is downhill, step, repeat



EACH STEP

- Start anywhere: slope 0, miss 1,092
- Gradient: which way the miss rises, and how fast
- Step against it: w ← w − learning rate × gradient
- Slope 0 → 4.6 → 8.3 → 11.2 → 13.5: miss 1,092 → 548

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Learning rate 0.1, standardised feature; dots are the real steps.

Everyday: walking downhill in fog to find the valley

THE STORY

You can’t see the bottom, only the slope under your feet. Step downhill, check again, repeat until it’s flat.

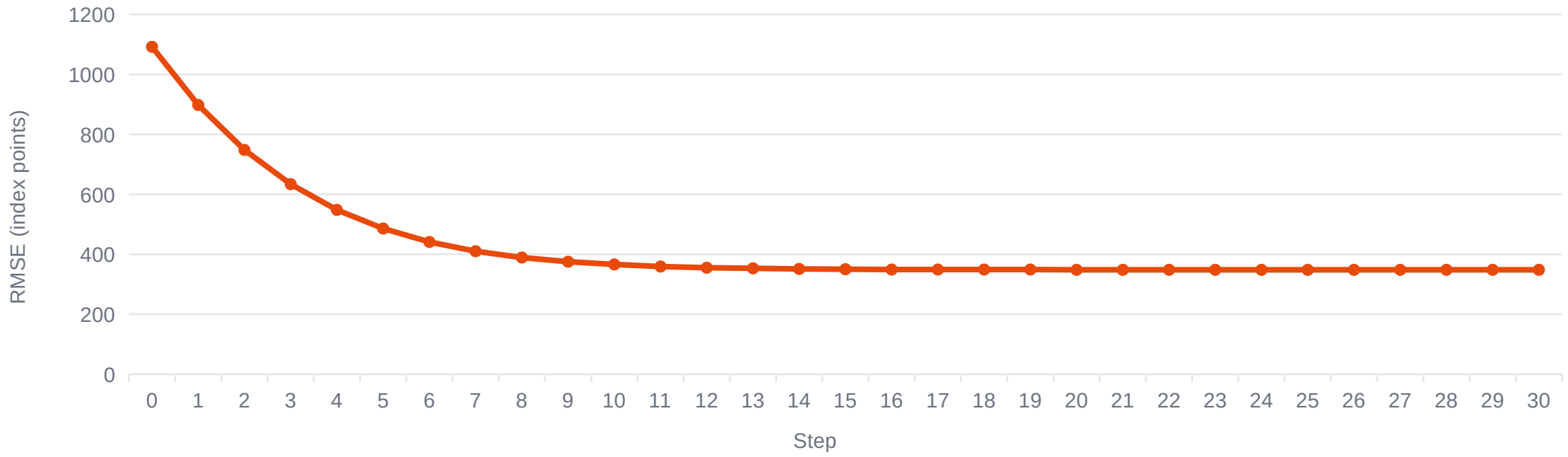
w: 0 → 2 → 3.5 → 4.5 → 5

Made-up numbers, for illustration

Hiking	In the model	Exam example
Where you stand	Current guess for w	w = 0
Slope underfoot	Gradient: which way error falls	Raise w
Step size	Learning rate	Small steps
Valley floor	Lowest error	w = 5

IN OUR FINANCE EXAMPLE **Our slope walked 0 → 4.6 → 8.3 → … → 22.9.**

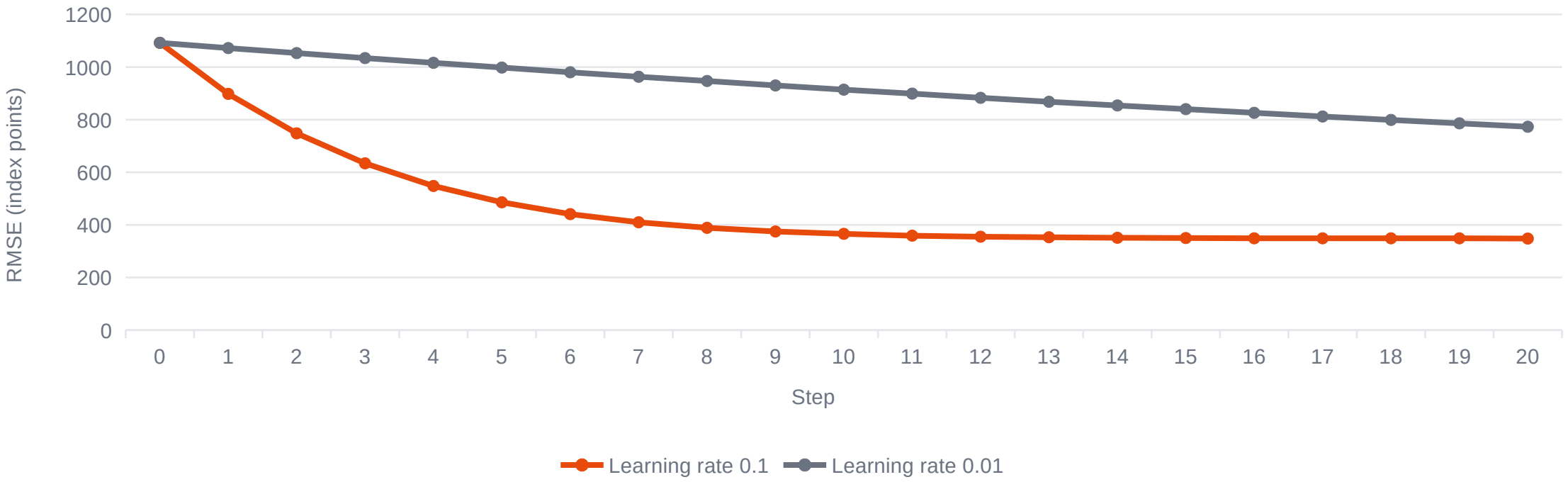
Thirty steps take the typical miss from 1,092 to 348 points



Big gains early, then a plateau at the least-squares answer.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Learning rate 0.1, standardised features.

At 0.01 training crawls; at 0.1 it converges in about 20 steps



Too large (1.05) diverges: the error grows to 6,969 by step 20.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Standardised features.

Everyday: setting the temperature in the shower

THE STORY

Tiny turns of the knob: you wait forever. Huge turns: scalding, then freezing. Medium turns: just right.

Too small = slow · Too big = overshoots

Made-up numbers, for illustration

Learning rate	Shower knob	What happens
Too small	Tiny turns	Still cold after 20 tries
About right	Medium turns	Right temperature, fast
Too big	Huge turns	Hot, cold, hotter: never settles

IN OUR FINANCE EXAMPLE **0.01 → 773 after 20 steps; 0.1 → 348; 1.05 blows up.**

Your loss curve goes up every step. What do you change?

CONTEXT Each step: w ← w − learning rate × gradient. With learning rate 1.05 on our 44 rows the error goes 1,092 → 1,190 → 1,299 and keeps climbing. With 0.1 it fell to 348.

A Run more steps; it will turn around eventually

B Lower the learning rate

C Add more features to the model

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WHY

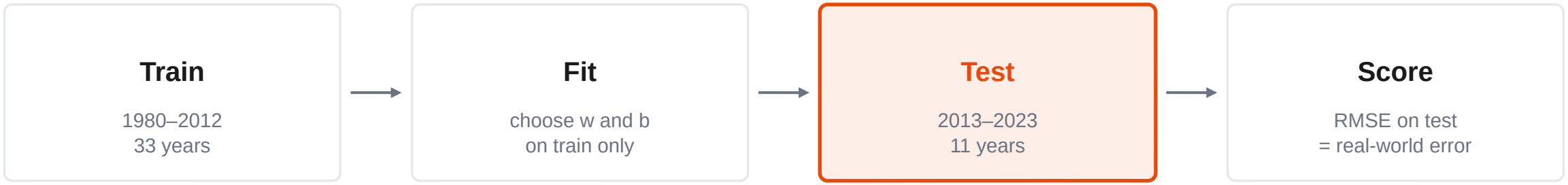
Each step overshoots the bottom of the valley. These are the first steps of the 1.05 run; at 0.1 the same data converges to 348.

06

Model evaluation

110–120 min · does the model survive 2013–2023, years it never saw? A backtest done right.

Fit on the past, score on years the model never saw



Split by time: the model must never see the years it is scored on.

Everyday: practice exams vs the real exam

THE STORY

Memorise the practice answer key and you ace the practice test, then fail the real one. That is overfitting.

Practice 100% · Real exam 55% → overfit

Made-up numbers, for illustration

Term	Meaning	Exam example
Training data	What you learn from	Last year’s practice tests
Test data	Never seen before	Friday’s final
Overfitting	Memorised, didn’t learn	100% practice, 55% final
Underfitting	Didn’t learn enough	60% on both

IN OUR FINANCE EXAMPLE **Train on 1980–2012, test on 2013–2023. The cubic aced the past and failed the test.**

Three models fitted on 1980–2012. Which do you ship to the PM?

CONTEXT All three were fitted on 1980–2012. Typical miss on those same training years: average only 473, straight line 252, cubic curve 214 index points. 2013–2023 is held back and not yet scored.

A The cubic: lowest error on the training years

B Can't tell yet: score them on 2013–2023 first

C The average: simplest is always best

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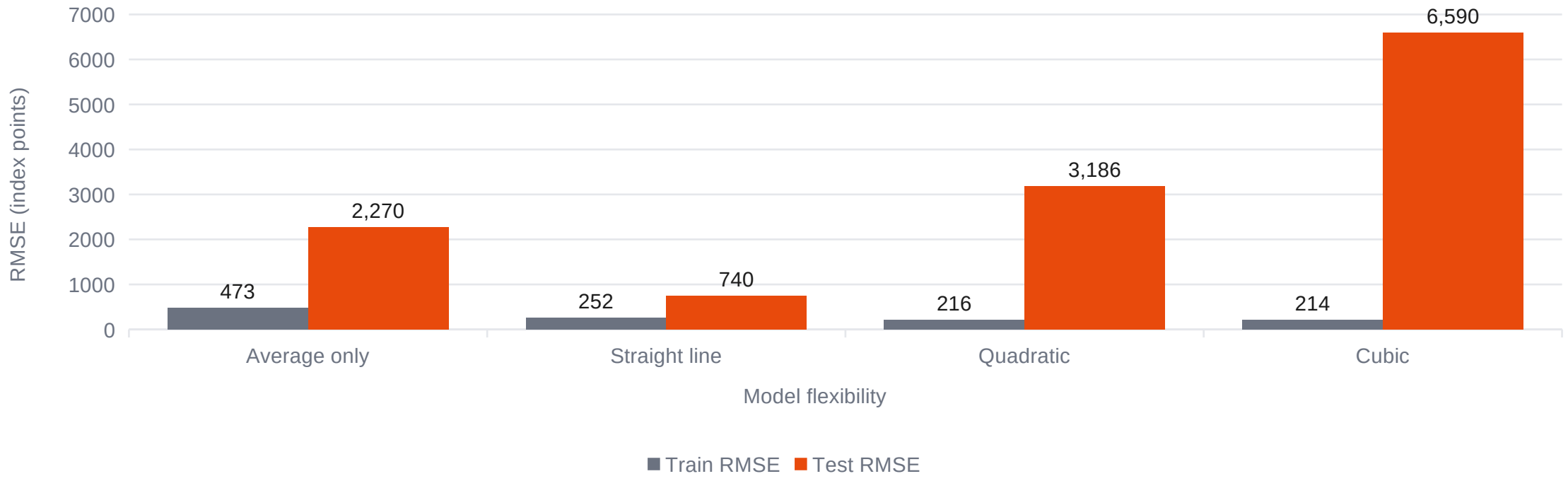
✓ **B. Can't tell yet: score them on 2013–2023 first**

C The average: simplest is always best

WHY

Training error always falls as a model gets more flexible. Only the held-out years tell you which one generalizes. Next slide.

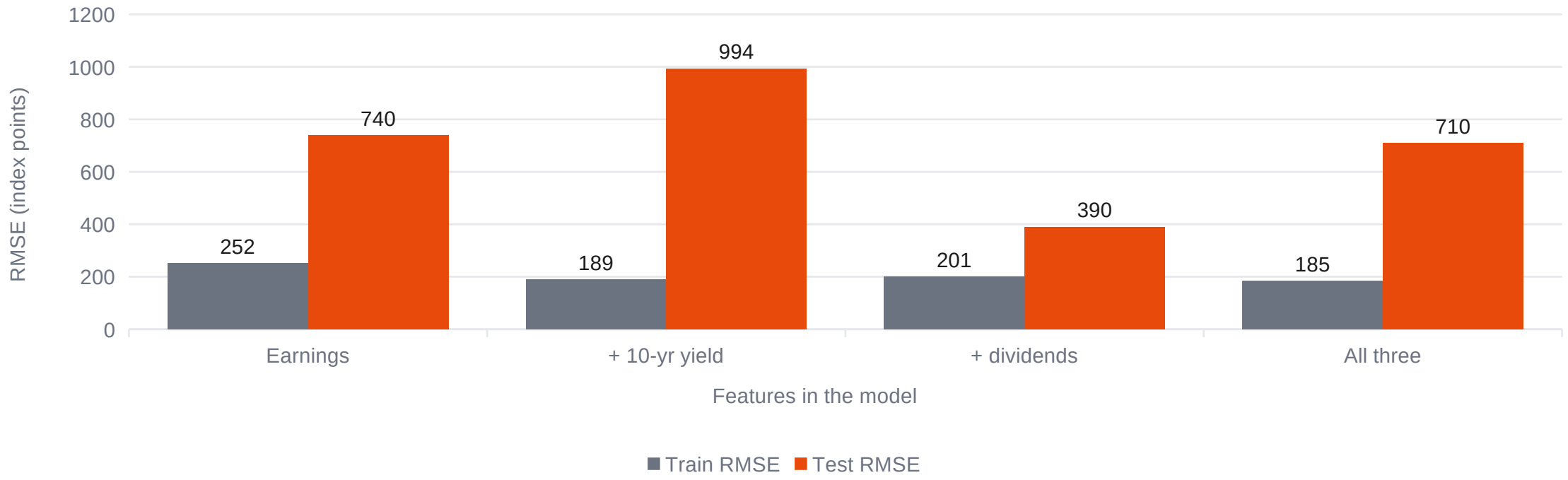
More flexible curves fit the past better and the future far worse



The straight line has the lowest test error: 740 points.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Polynomial fits of index on earnings.

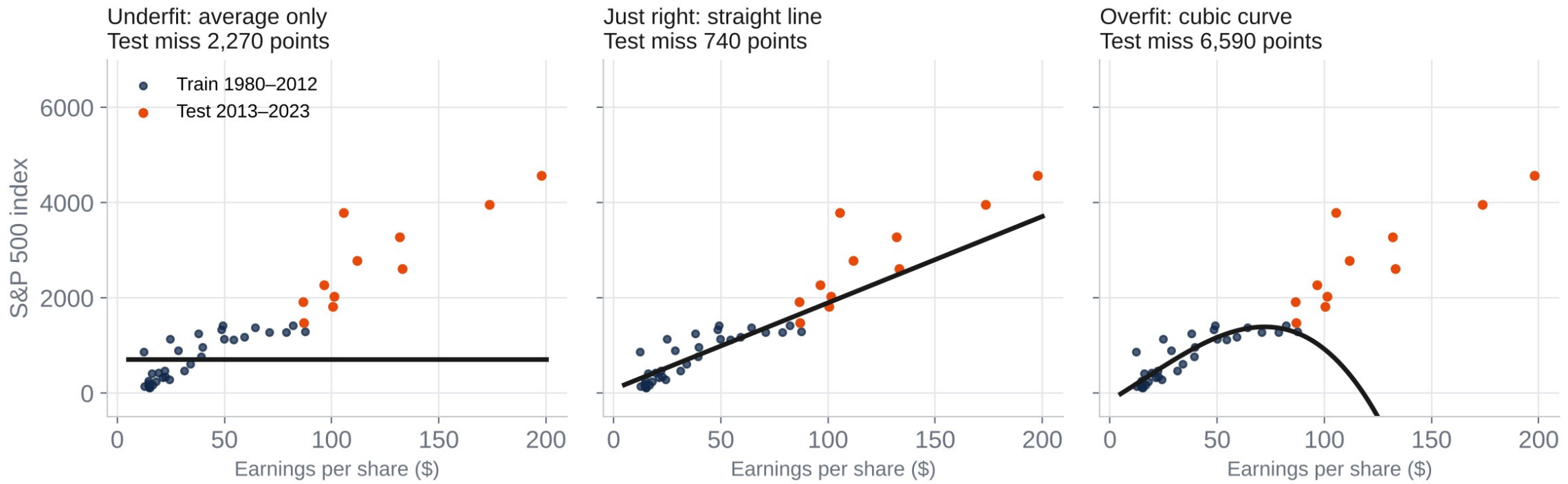
Adding the 10-year yield fit history better and 2013–2023 worse



Earnings + dividends generalized best: 390. The yield helped on history only.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Long rate = 10-year Treasury yield.

A good model generalizes: it works on data it has never seen



Ship the simplest model with the lowest test error: here, the straight line.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Fitted on 1980–2012 (navy); orange = 2013–2023 test years.

07

Supervised learning: more models

Beyond the plain line: ridge and lasso, classifiers, and ensembles of trees, on real market data.

Three ways to fit the line: what each one minimises

Linear

Lowest prediction error

Minimise: squared error

No penalty; weights can run wild

Lasso

Lowest error plus absolute-weight penalty

Minimise: squared error + $\lambda \cdot \Sigma |w|$

Sets weak weights to exactly zero

Ridge

Lowest error plus squared-weight penalty

Minimise: squared error + $\lambda \cdot \Sigma w^2$

Shrinks all weights, drops none

λ sets how hard you push. Next slide: how to choose it.

Everyday: too many inputs, some of them useless

THE STORY

Predict the score from hours, sleep, coffee and shoe size. Shoe size is noise. A penalty stops the model leaning on it.

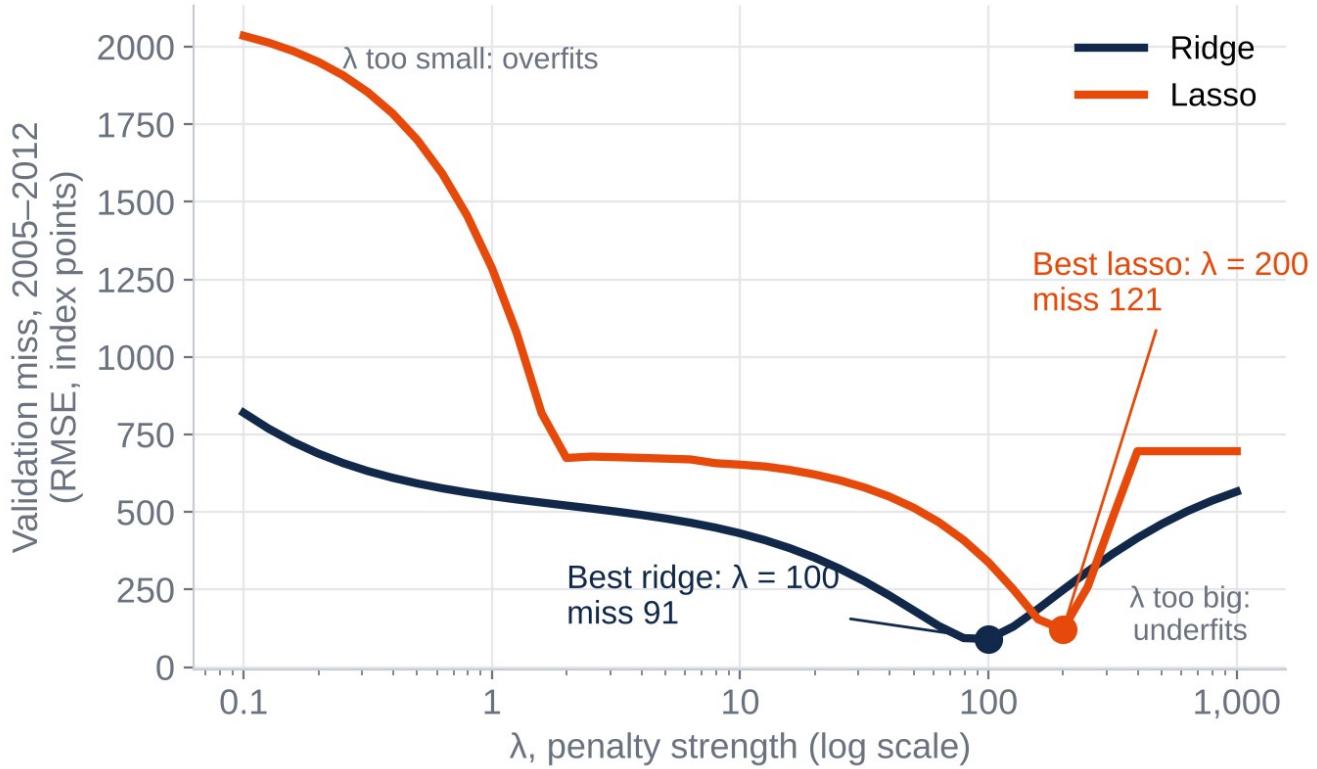
Lasso: shoe size → 0 · Ridge: all shrink

Made-up numbers, for illustration

Feature	Plain	Ridge	Lasso
Hours studied	5.0	4.6	4.8
Hours slept	2.0	1.8	1.5
Cups of coffee	1.2	0.6	0
Shoe size	−3.1	−0.4	0

IN OUR FINANCE EXAMPLE **Lasso kept earnings and CPI and dropped the other four.**

Pick λ: try many, keep the one that wins on held-out years



THE RECIPE

- Fit on 1980–2004, once for each λ
- Score each fit on 2005–2012, the validation years
- Keep the lowest miss: ridge λ = 100, lasso λ = 200
- Refit on 1980–2012, then test 2013–2023 once

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Six features, standardised. Validation RMSE in index points.

Everyday: how strict should the penalty be?

THE STORY

Try several strictness levels on practice test 1, check each on practice test 2, keep the best. Never peek at the final.

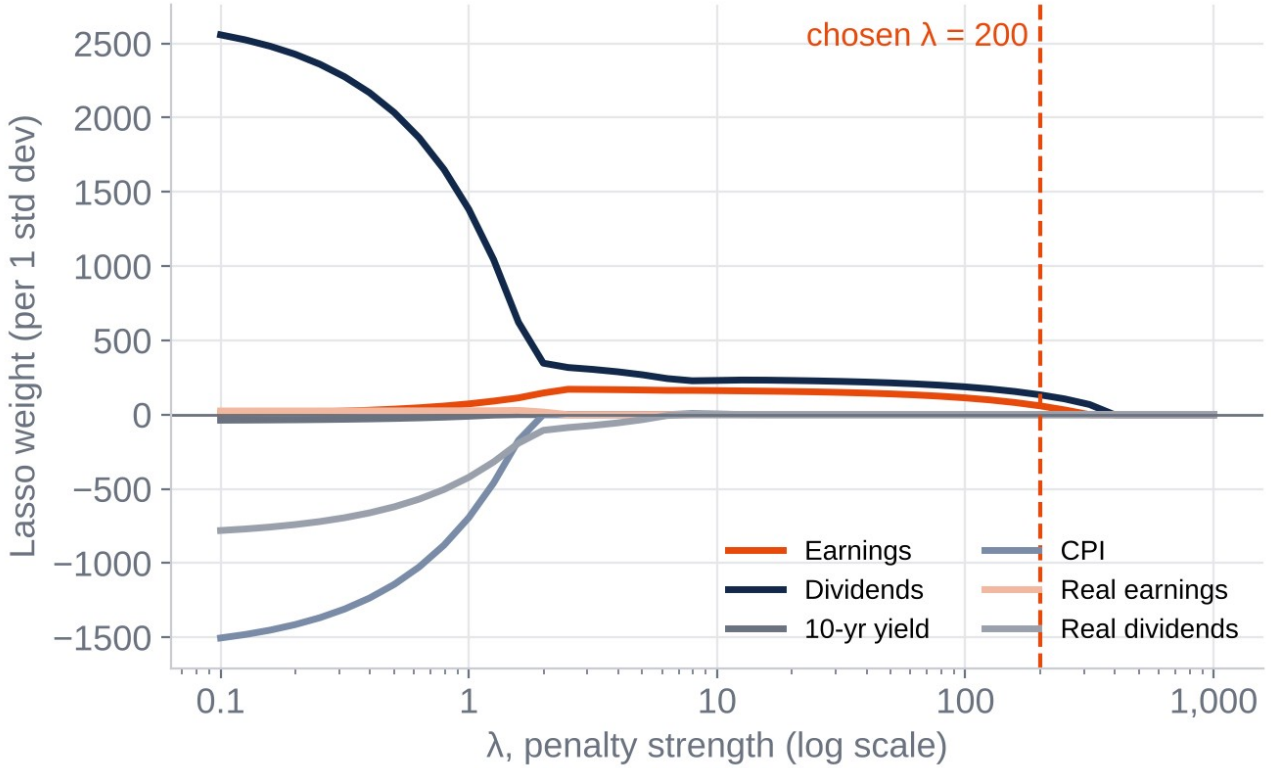
Keep λ = 10: best on practice test 2

Made-up numbers, for illustration

λ	What happens	Practice test 2
0	No penalty: uses shoe size, overfits	68
10	Useless inputs dropped	84
1,000	Every weight ≈ 0: guesses the average	61

IN OUR FINANCE EXAMPLE **Ridge λ = 100 and lasso λ ≈ 200, both chosen on 2005–2012.**

Raise λ and lasso shrinks weights, dropping features one by one

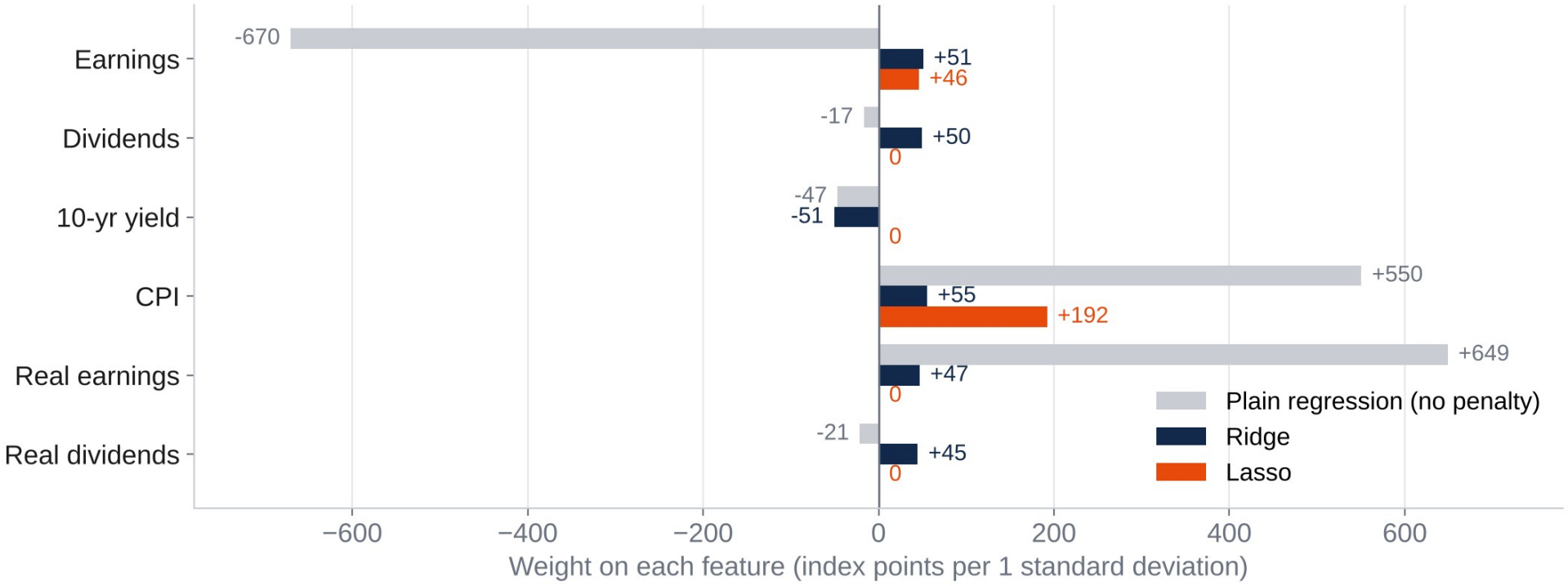


READING THE PATH

- λ near 0: plain regression, weights in the thousands
- λ = 10: three features left
- λ = 50 to 200: two left, earnings and dividends
- λ = 500: none left, just the average

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Fit on 1980–2004; six features, standardised.

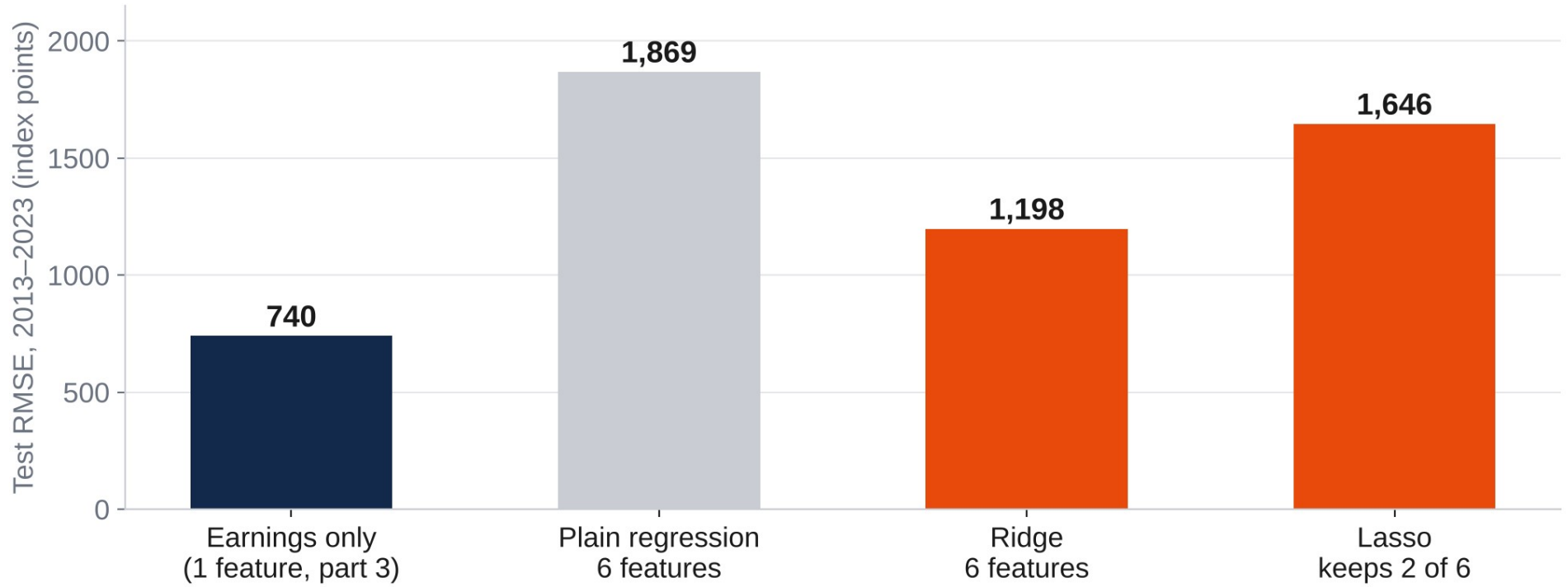
Plain regression's weights go haywire; penalties tame them



No penalty: −670 on earnings, +649 on real earnings. Ridge: every weight near ±50.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Fit 1980–2012, features standardised; λ chosen on 2005–2012.

Penalties cut the test miss, but one good feature still wins



Six features, no penalty: 1,869. Ridge: 1,198. The one-feature line from part 3: 740.

Source: Shiller S&P 500 data (github.com/datasets/s-and-p-500), January of each year, 1980–2023, n = 44. Train 1980–2012, test 2013–2023.

Why pick λ on 2005–2012 instead of on the 2013–2023 test years?

CONTEXT λ is the penalty strength we have to choose. Our recipe: fit on 1980–2004 → choose λ by the lowest miss on 2005–2012 → score once on 2013–2023, the test years that stand in for the future.

A Choosing λ on the test years leaks the answer into the model

B Older years are more accurate

C It makes no difference which years you use

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✓ **A. Choosing λ on the test years leaks the answer into the model**

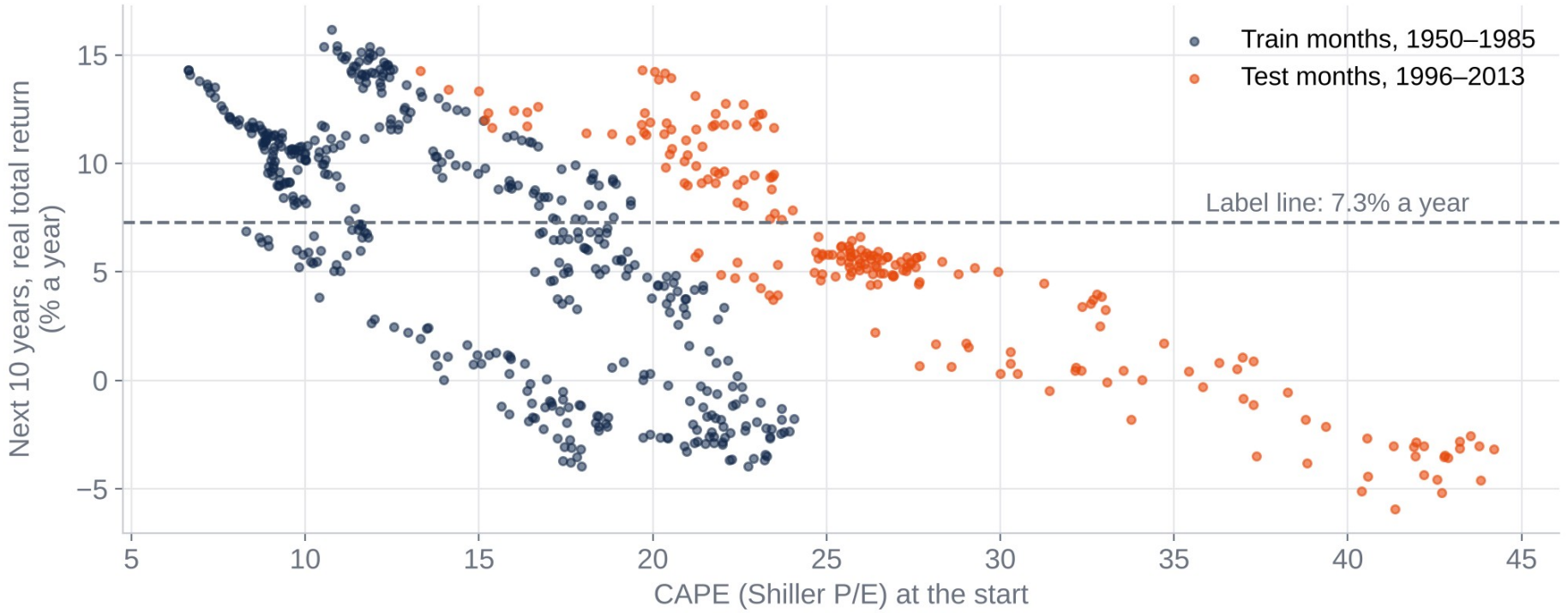
B Older years are more accurate

C It makes no difference which years you use

WHY

Any choice made by looking at test results makes the test score too rosy. The test must be touched once, at the end.

Will the next 10 years beat 7.3% a year after inflation?

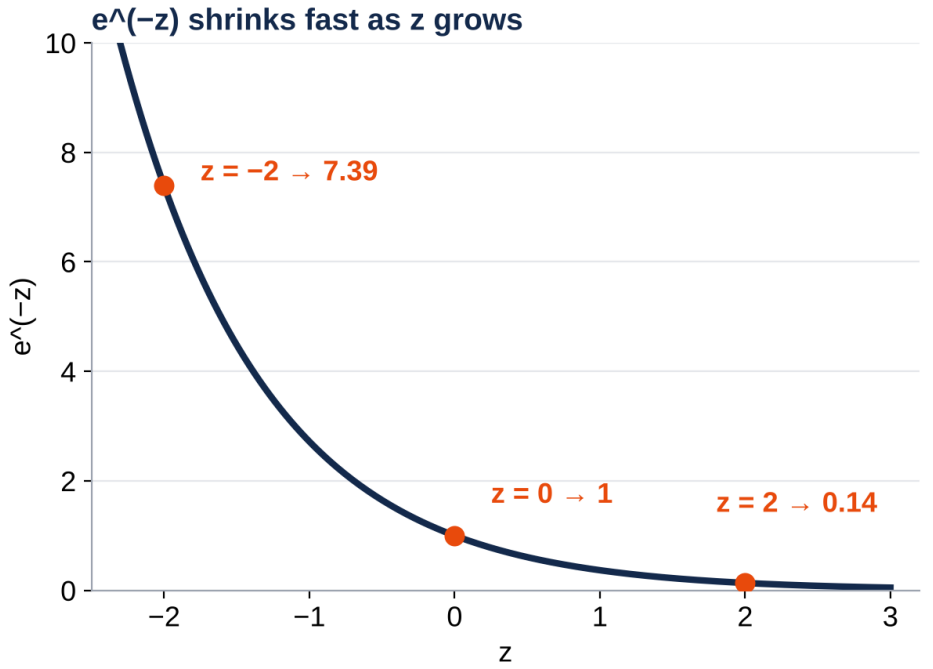


Expensive starts were followed by weaker decades. Test years start pricier than any training year.

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Label line = median of the training years.

An exponent says how many times to multiply a number by itself

2^3	$= 2 \times 2 \times 2 = 8$	The small raised number is the exponent
8^2	$= 8 \times 8 = 64$	Power 2 = “squared”: how MSE scores a miss of 8
$2^0 = 1$	$2^{-1} = ½$	Zero gives 1; a minus sign means divide
$e \approx 2.718$		A special constant; e^(−z) appears in the sigmoid



Big z → e^(−z) near 0 → probability near 100%. Very negative z → e^(−z) huge → probability near 0%.

Score, squeeze, decide: logistic regression in three steps

1 Score

$$z = b + w_1 \cdot \text{CAPE} + w_2 \cdot \text{yield}$$

A weighted sum, exactly like linear regression. Can be any number.

2 Squeeze

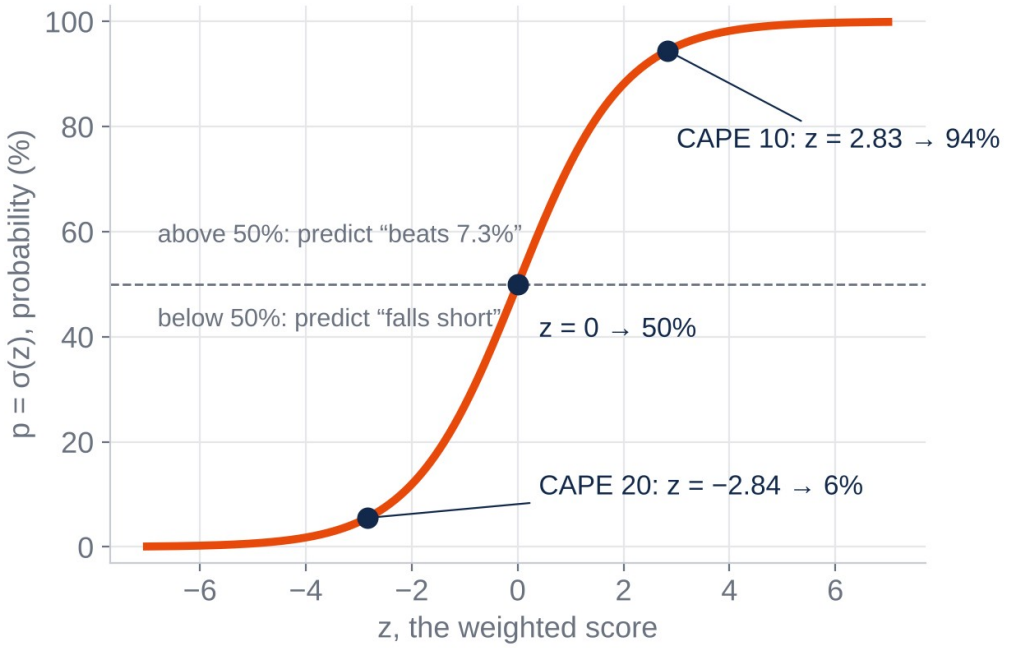
$$p = \sigma(z) = 1 / (1 + e^{(-z)})$$

The sigmoid turns any z into a probability between 0 and 1.

3 Decide

$$p \geq 0.5 \rightarrow \text{“beats 7.3\%”}$$

z = 0 is the 50% line. Lower the cut-off to catch more.



Points on the curve are the fitted model's real outputs (next slide). Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends.

Everyday: the chance of passing from hours studied

THE STORY

Score: z = −4 + 1 × hours. Squeeze z through the sigmoid into a 0–100% chance. Decide: pass if above 50%.

$$p = 1 \div (1 + e^{(-z)})$$

Made-up numbers, for illustration

Hours	z = −4 + hours	Chance of passing
2	−2	12% → fail
4	0	50% → coin flip
6	+2	88% → pass

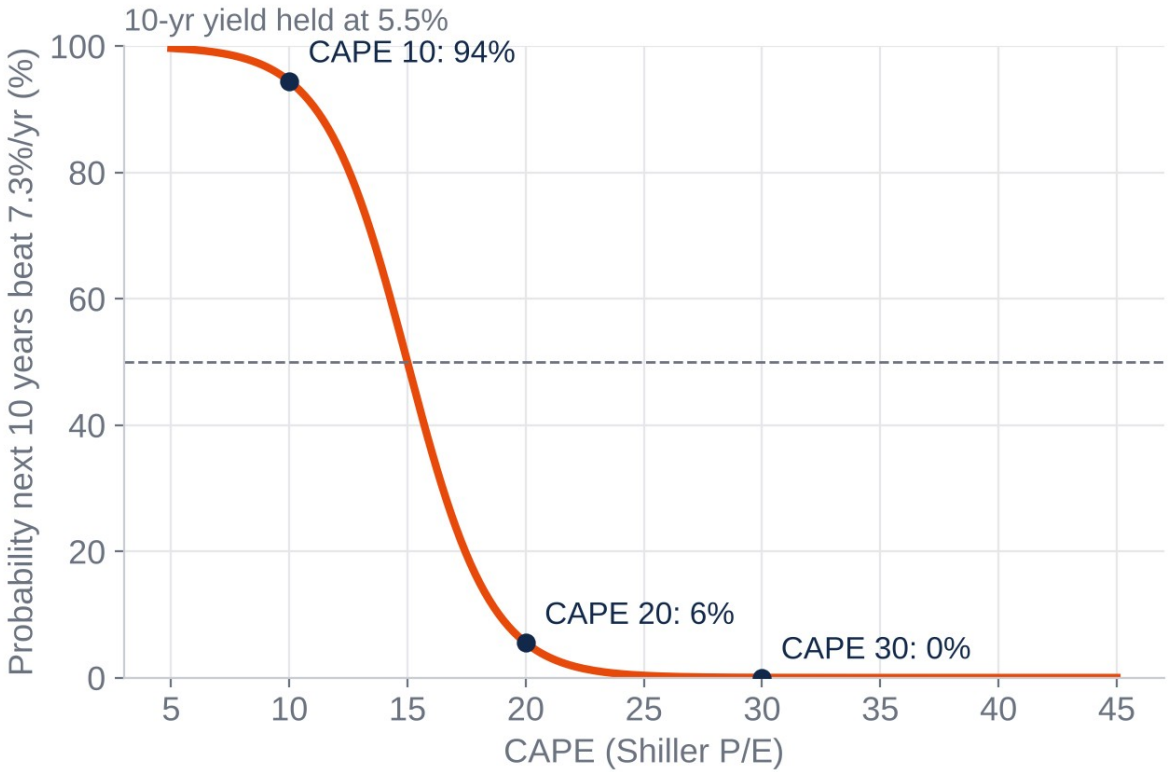
IN OUR FINANCE EXAMPLE **z = 10.33 − 0.567·CAPE − 0.332·yield. CAPE 10 → 94%.**

Plugging real numbers into the fitted model

1. Fitted on 1950–1985: z = 10.33 − 0.567 · CAPE − 0.332 · yield
2. CAPE 10, yield 5.49%: z = 10.33 − 5.67 − 1.82 = 2.83
3. Squeeze: p = 1 / (1 + e^(−2.83)) = 0.94 → 94%: predict ‘beats’
4. CAPE 20, same yield: z = −2.84 → p = 6%, predict ‘falls short’

Weights learned by minimising log loss with gradient descent, the part 5 loop.

Logistic regression turns a weighted sum into a probability



HOW IT WORKS

- Weighted sum of CAPE and yield, squeezed into 0–100%
- CAPE 10 → 94%. CAPE 20 → 6%
- Above 50%: predict ‘beats 7.3%’
- Test accuracy: 70.9% vs 67.6% for guessing

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Curve holds the 10-yr yield at its training median, 5.5%.

The logistic model gives z = 0 for a month. What does it predict?

CONTEXT z = the weighted score. p = σ(z) = 1 / (1 + e^(−z)). Predict ‘beats 7.3%’ when p ≥ 0.5. Remember $e^0 = 1$.

A p = 0%: falls short

B p = 50%: right on the boundary, a coin flip

C p = 100%: beats 7.3%

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A p = 0%: falls short

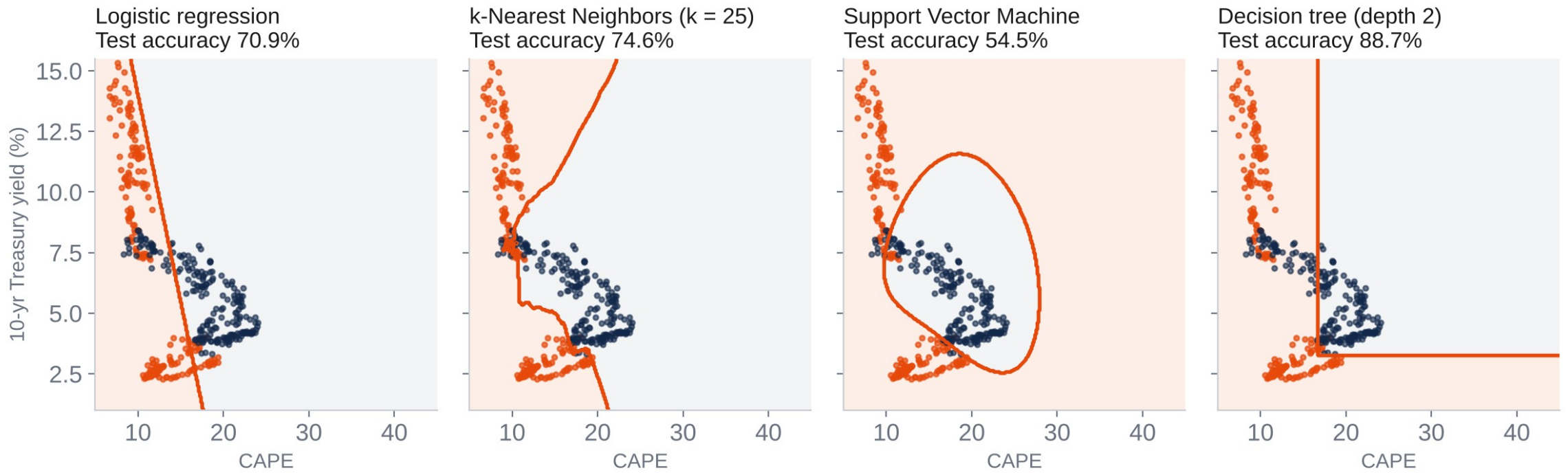
✓ **B. p = 50%: right on the boundary, a coin flip**

C p = 100%: beats 7.3%

WHY

$e^0 = 1$, so p = 1 / (1 + 1) = 0.5. z = 0 is exactly the decision boundary.

Same data, four classifiers, four different boundaries



Orange area: predicts ‘beats 7.3%’. Dots: training months (orange beat, navy fell short).

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Train 1950–1985, test 1996–2013.

Everyday: four ways to call pass or fail

THE STORY

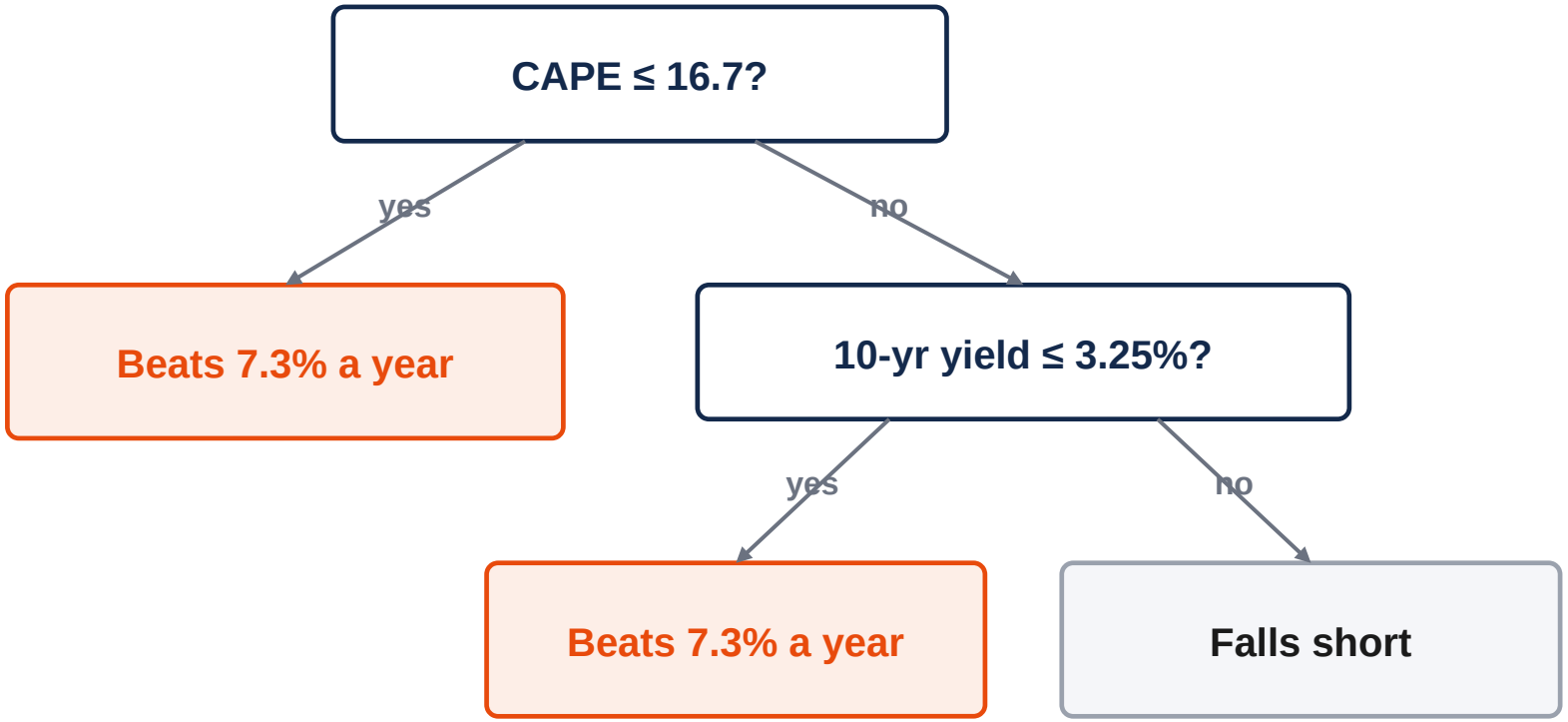
Same students, same two inputs: hours studied and hours slept. Each method draws the pass/fail line differently.

Made-up numbers, for illustration

Method	How it decides	Exam example
Logistic regression	Weighted score → probability	6 h → 88% pass
k-nearest neighbours	Ask the most similar past students	4 of 5 lookalikes passed
SVM	Widest gap between pass and fail	Gap near 4 hours
Decision tree	Yes/no questions	Studied ≥ 4 h? Slept ≥ 6 h?

IN OUR FINANCE EXAMPLE **Same idea on CAPE and the 10-year yield: the tree scored 88.7%.**

A decision tree is a flowchart learned from the data



WHY TEAMS LIKE IT

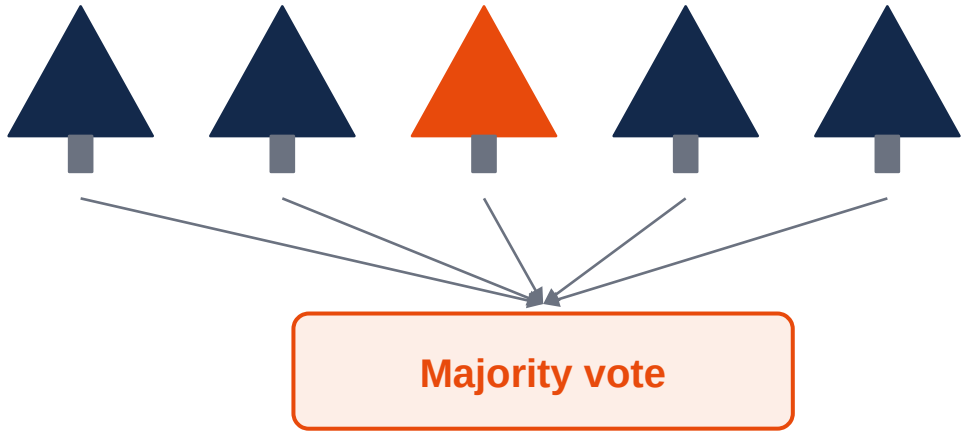
- Every prediction is a rule a PM can read
- Test accuracy: 88.7%
- Too deep and it memorises noise

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Thresholds are the splits learned on 1950–1985 (depth 2).

Ensembles: many trees, combined, beat one tree

Random forest

Many trees in parallel, each on a random sample. They vote.



Majority vote

Gradient-boosted trees

Small trees in sequence; each fixes the last one’s errors.



Sum of all trees

The cost: 500 trees are harder to explain than one. Feature importance helps.

Everyday: ask many tutors instead of one

THE STORY

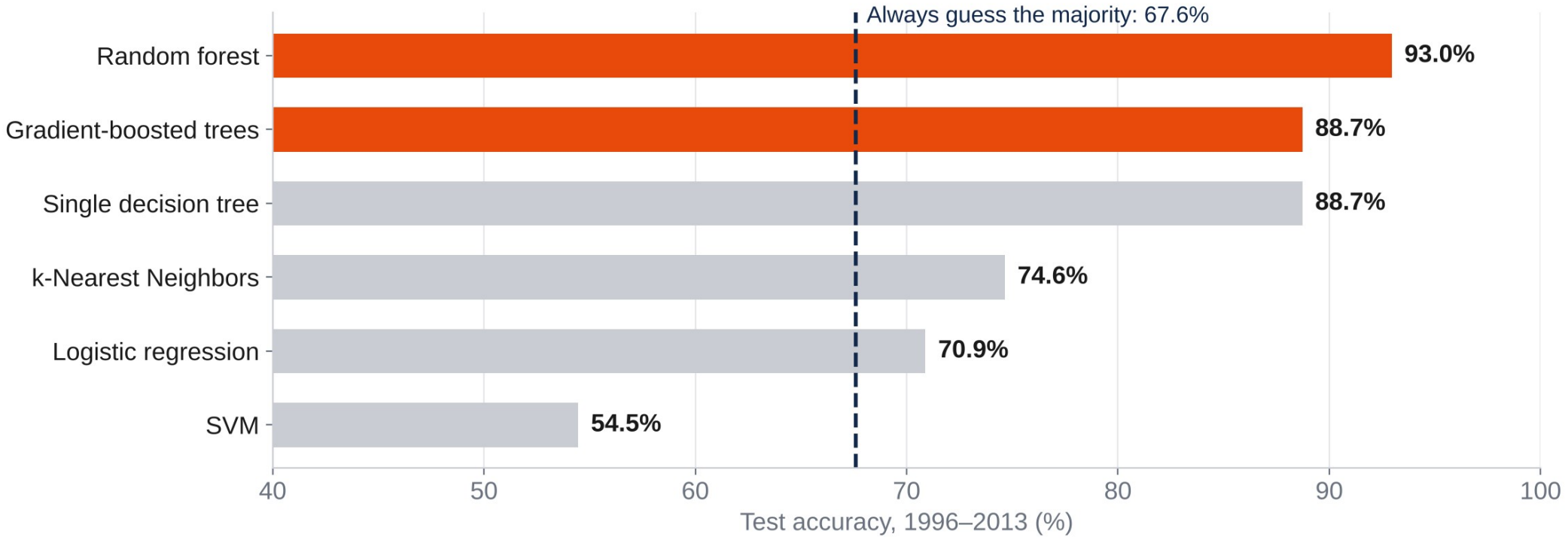
One tutor can be quirky. Many tutors who each studied different material, voting together, are harder to fool.

Made-up numbers, for illustration

Method	Everyday version	Result
One tree	One tutor’s opinion	Can be quirky
Random forest	100 tutors, each saw different problems; they vote	70 say pass → pass
Gradient boosting	Each new tutor fixes the last one’s mistakes	Errors shrink each round

IN OUR FINANCE EXAMPLE **Random forest 93% vs one tree 88.7% on the 10-year question.**

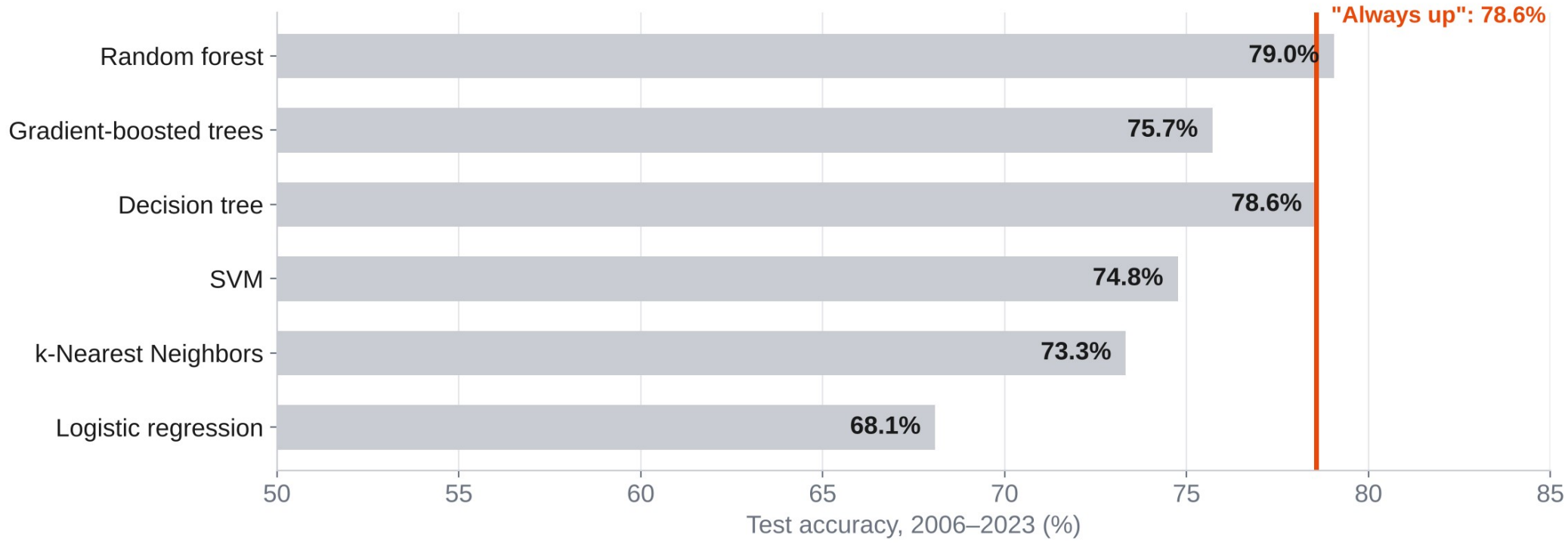
Random forest wins: 93% accuracy vs 68% for guessing



Trees and ensembles handled the expensive test years; SVM did worse than guessing.

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Test months 1996–2013; overlapping 10-year windows, so treat as illustration.

Next 12 months up or down? No model beats always saying ‘up’



The S&P 500 rose in 79% of test months. Beat the base rate, not 50%.

Source: Shiller S&P 500 monthly data (github.com/datasets/s-and-p-500). Real total return = real price plus real dividends. Train 1950–2004, test 2006–2023; features CAPE, 10-yr yield, dividend yield, 12-mo momentum, inflation.

Everyday: if 90% pass, “everyone passes” scores 90%

THE STORY

A rule that says “pass” for every student is right 9 times out of 10 without learning anything.

Always compare with the dumbest sensible rule

Made-up numbers, for illustration

Rule	Accuracy	Verdict
Always say “pass”	90%	The bar to beat
A fancy model	88%	Worse than doing nothing

IN OUR FINANCE EXAMPLE **Always “up”: 78.6%. Gradient boosting: 75.7%.**

SUPERVISED LEARNING CHECKPOINT · TAKE 30 SECONDS

A model calls next year’s market direction right 75% of the time. Is that good?

> **CONTEXT** In the 2006–2023 test months, the S&P 500 was higher a year later 79% of the time. A rule that just says ‘up’ every month needs no model at all.

A Yes: 75% is far above a coin flip

B No: always saying ‘up’ would score 79%

C Can’t say without more features

A model calls next year’s market direction right 75% of the time. Is that good?

CONTEXT In the 2006–2023 test months, the S&P 500 was higher a year later 79% of the time. A rule that just says ‘up’ every month needs no model at all.

A Yes: 75% is far above a coin flip

✓ **B. No: always saying ‘up’ would score 79%**

C Can’t say without more features

WHY

Compare with the base rate, not 50%. A model that loses to ‘always up’ adds nothing.

What to take away

01 ML learns the rule from data: supervised, unsupervised, reinforcement.

02 Linear regression $\hat{y} = w \cdot x + b$; the loss (RMSE 348) scores its misses.

03 Gradient descent walks downhill; test on unseen years to catch overfitting.

04 Ridge and lasso tame overfitting; pick λ on held-out years.

05 Classify with probabilities; ensembles won here, but check the base rate.

BEFORE NEXT FRIDAY

Rebuild today’s loop yourself in the notebook

- Supervised: fit the line on 1980–2012 (sp500_shiller_january_1980_2023.csv); confirm test RMSE ≈ 740.
- Gradient descent: try learning rates 0.01, 0.1, 1.05. Plot the loss curves.
- Stretch: add dividends and the 10-year yield. Does test error fall?

Next: ML Foundations II, Fri Oct 23.

Questions?

Next: Chapter 5, ML Foundations II · Friday, Oct 23.

GO DEEPER

Data: Shiller S&P 500 dataset — github.com/datasets/s-and-p-500

James et al., An Introduction to Statistical Learning — ch. 3, 4, 6, 8

Google Machine Learning Crash Course — Linear regression, Loss, Gradient descent

Today's chapter

BEFORE YOU LEAVE

Scan to mark your attendance

Attendance · Workshop 4

Fri, Oct 16, 2026 · 1:30–3:30 pm · 3007 BIF

1. Point your phone camera at the code.
2. Fill in the short attendance form.
3. Attend 3 sessions to earn credit for the workshop.